

Inverse Manipulator Kinematics

Chapter 4

Inverse Manipulator Kinematics

Forward vs. Inverse Kinematics

Forward (Direct) Kinematics

Given:

Joint angles and links geometry

Compute:

Position and orientation of the end effector relative to the base frame

$$f(\theta) = {}^B_T T = {}^0_N T$$

Inverse Kinematics

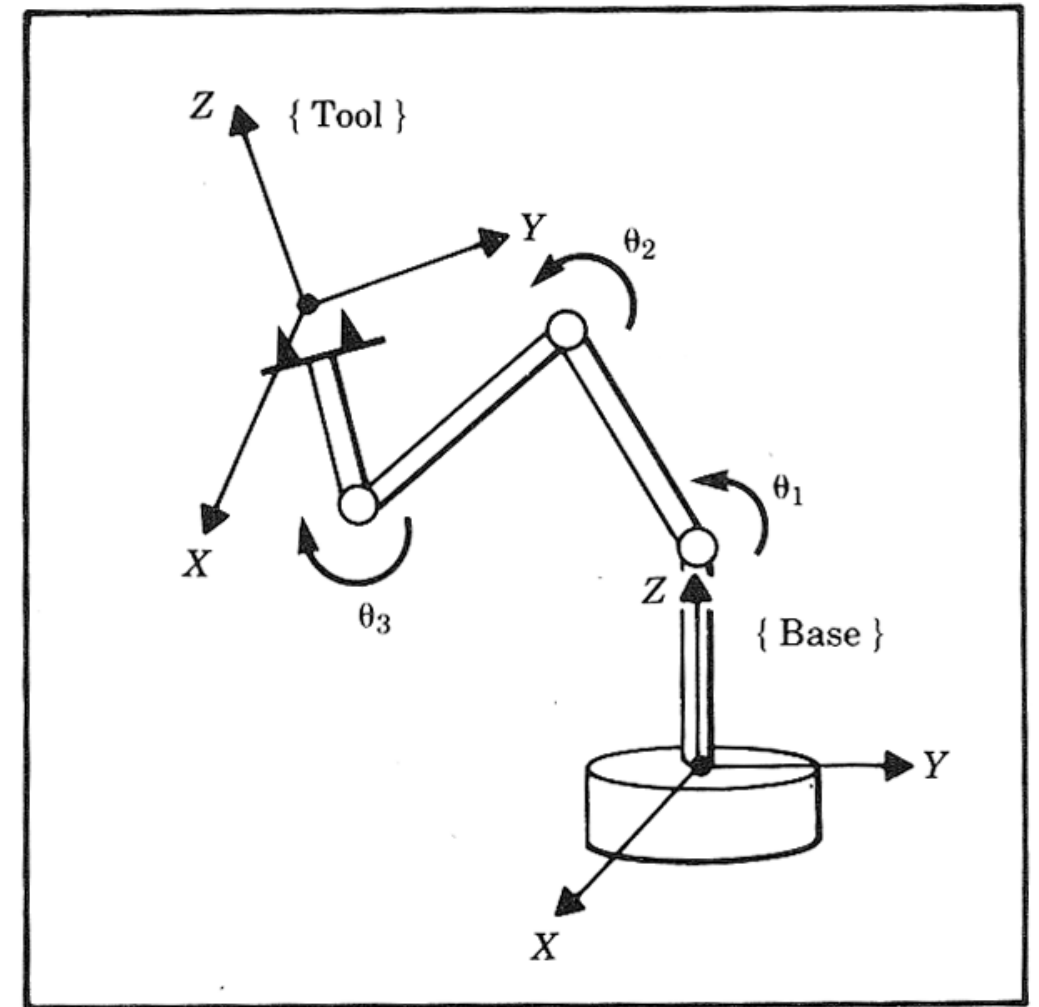
Given:

Position and orientation of the end effector relative to the base frame

Compute:

All possible sets of joint angles which could be used to attain the given position and orientation of the end effector

$$\theta = f^{-1}({}^B_T T) = f^{-1}({}^0_N T)$$



Inverse Manipulator Kinematics

Solvability - PUMA 560

Given : PUMA 560 - 6 DOF, 0_6T

Solve: $\theta_1 \cdots \theta_6$

$${}^0_6T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T {}^5_6T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Total Number of Equations: 12

Independent Equations: 3 - Rotation Matrix
3 - Position Vector

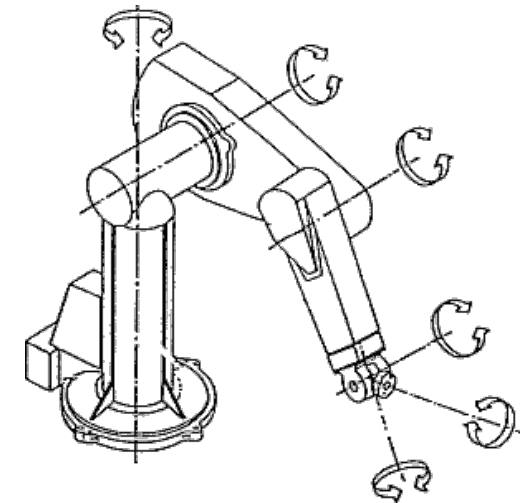
Type of Equations: Non-linear

$$\begin{aligned} r_{11} &= c_1 [c_{23}(c_4c_5c_6 - s_4s_6) - s_{23}s_5c_6] + s_1(s_4c_5c_6 + c_4s_6), \\ r_{21} &= s_1 [c_{23}(c_4c_5c_6 - s_4s_6) - s_{23}s_5c_6] - c_1(s_4c_5c_6 + c_4s_6), \\ r_{31} &= -s_{23}(c_4c_5c_6 - s_4s_6) - c_{23}s_5c_6, \end{aligned}$$

$$\begin{aligned} r_{12} &= c_1 [c_{23}(-c_4c_5s_6 - s_4c_6) + s_{23}s_5s_6] + s_1(c_4c_6 - s_4c_5s_6), \\ r_{22} &= s_1 [c_{23}(-c_4c_5s_6 - s_4c_6) + s_{23}s_5s_6] - c_1(c_4c_6 - s_4c_5s_6), \\ r_{32} &= -s_{23}(-c_4c_5s_6 - s_4c_6) + c_{23}s_5s_6, \end{aligned}$$

$$\begin{aligned} r_{13} &= -c_1(c_{23}c_4s_5 + s_{23}c_5) - s_1s_4s_5, \\ r_{23} &= -s_1(c_{23}c_4s_5 + s_{23}c_5) + c_1s_4s_5, \\ r_{33} &= s_{23}c_4s_5 - c_{23}c_5, \end{aligned}$$

$$\begin{aligned} p_x &= c_1 [a_2c_2 + a_3c_{23} - d_4s_{23}] - d_3s_1, \\ p_y &= s_1 [a_2c_2 + a_3c_{23} - d_4s_{23}] + d_3c_1, \\ p_z &= -a_3s_{23} - a_2s_2 - d_4c_{23}. \end{aligned}$$



Inverse Manipulator Kinematics

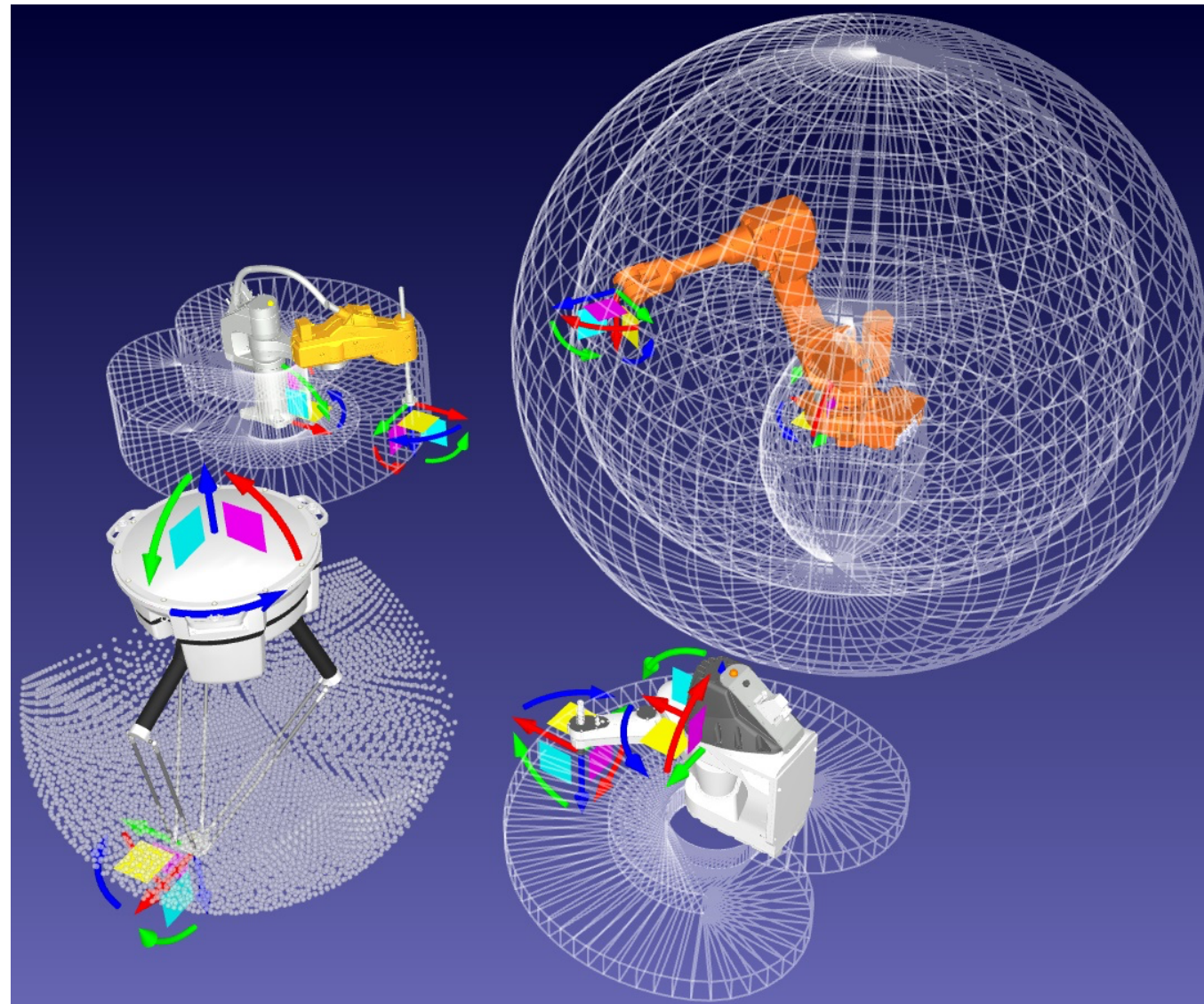
Solvability

- **Existence of Solutions**
- **Multiple Solutions**
- **Method of solutions**

Inverse Manipulator Kinematics

Workspace

Workspace - The volume of space that the end-effector can reach



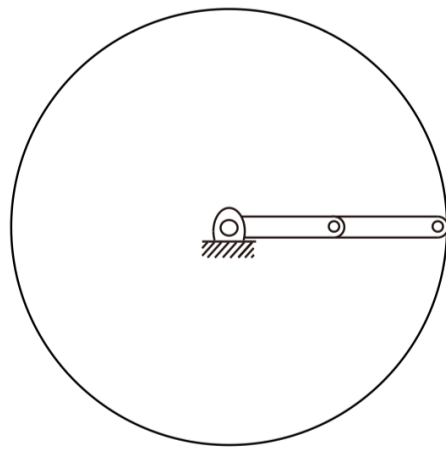
Workspace - The volume of space that the end-effector can reach



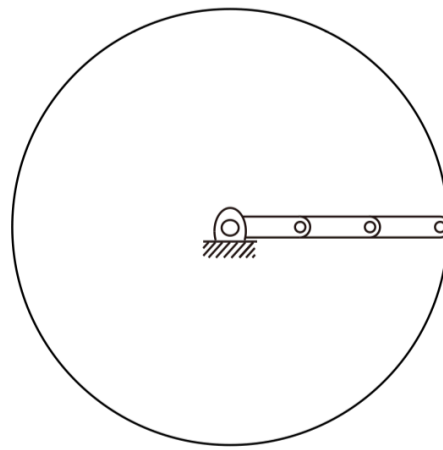
Inverse Manipulator Kinematics

Workspace

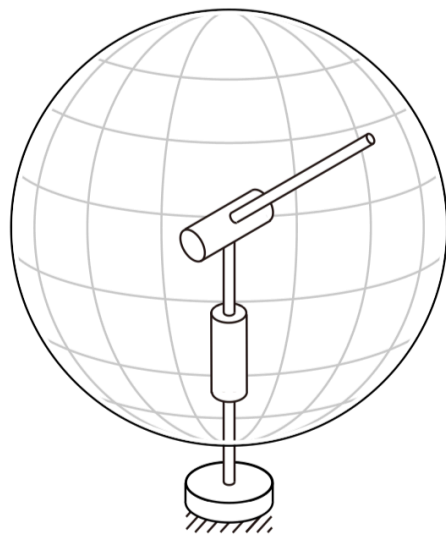
Workspace - The volume of space that the end-effector can reach



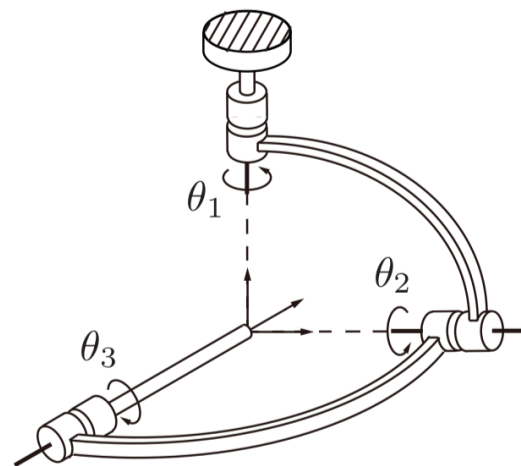
(a)



(b)



(c)



(d)

- (a) a planar 2R open chain
- (b) a planar 3R open chain
- (c) a spherical 2R open chain
- (d) a 3R orienting mechanism

Inverse Manipulator Kinematics

Solvability - Existence of Solution

- For a solution to exist, ${}^0_N T$ must be in the **workspace** of the manipulator
- **Workspace** - Definitions
 - **Reachable Workspace (RW):**
The volume of the space which every point can be reach by the end effector in at least one orientation
 - **Dexterous Workspace (DW):**
The volume of the space which every point can be reach by the end effector in all possible orientations

$$DW \subset RW$$

Inverse Manipulator Kinematics

Workspace - Cartesian

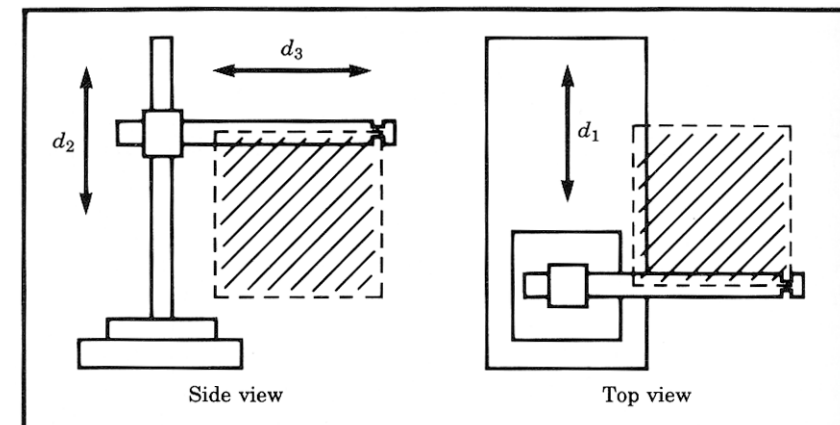
Joint 1 - Prismatic

Joint 2 - Prismatic

Joint 3 - Prismatic

Reachable Workspace?

Dexterous Workspace?



Inverse Manipulator Kinematics

Workspace - Cylindrical

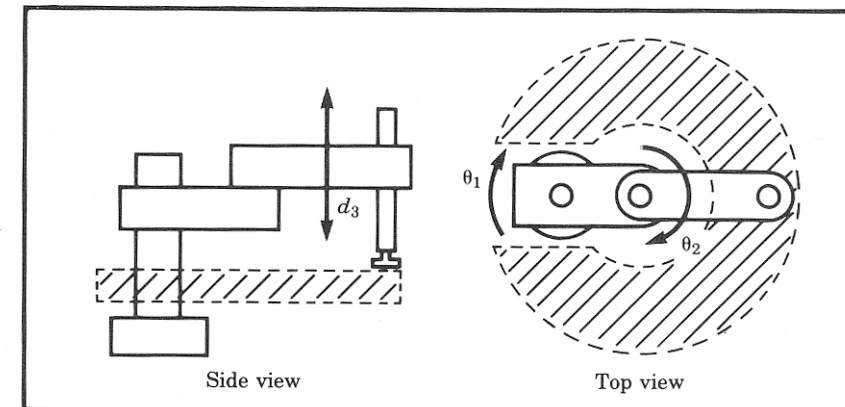
Joint 1 - Revolute

Joint 2 - Revolute

Joint 3 - Prismatic

Reachable Workspace?

Dexterous Workspace?



Inverse Manipulator Kinematics

Workspace - Spherical

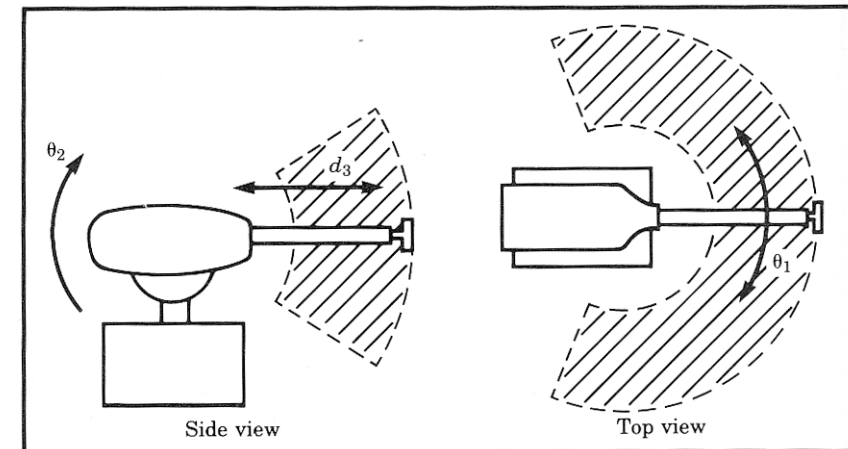
Joint 1 - Revolute (intersects with 2)

Joint 2 - Revolute (intersects with 1)

Joint 3 - Prismatic

Reachable Workspace?

Dexterous Workspace?



Inverse Manipulator Kinematics

Workspace - Cylindrical

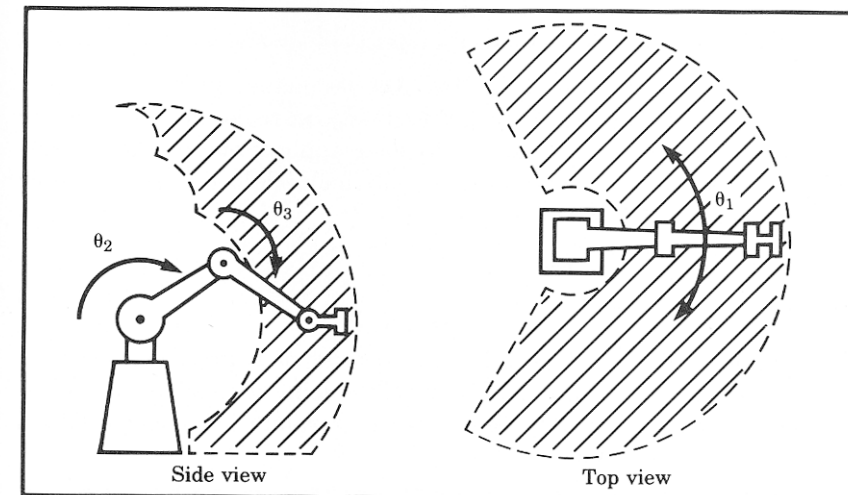
Joint 1 - Revolute

Joint 2 - Revolute

Joint 3 - Revolute

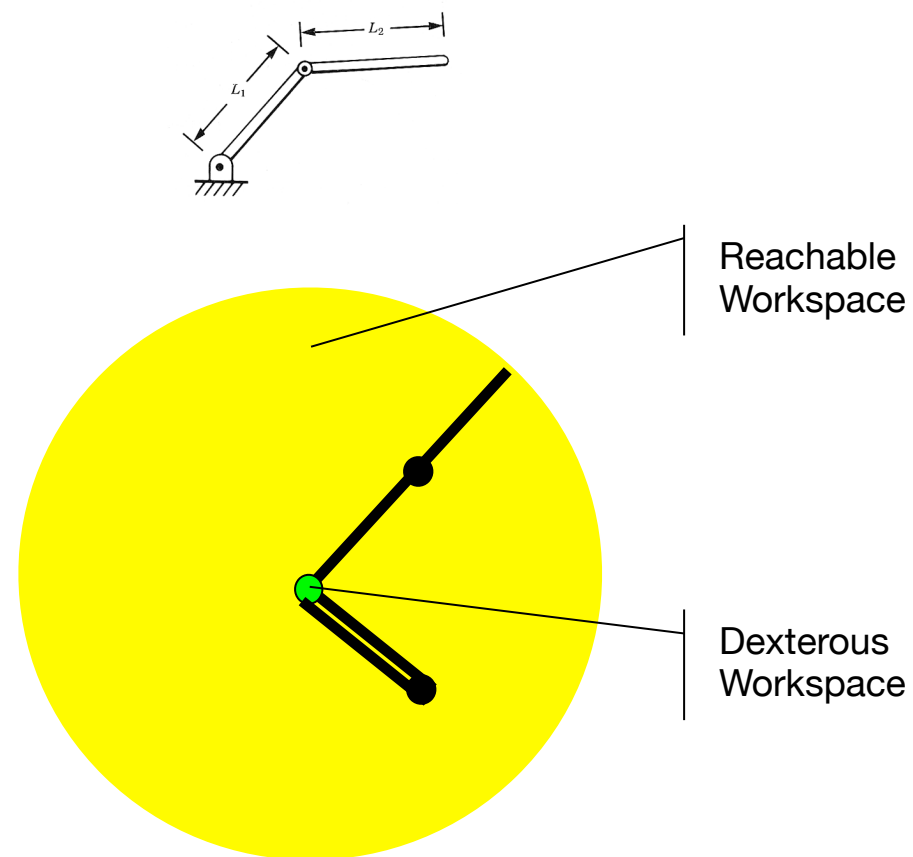
Reachable Workspace?

Dexterous Workspace?



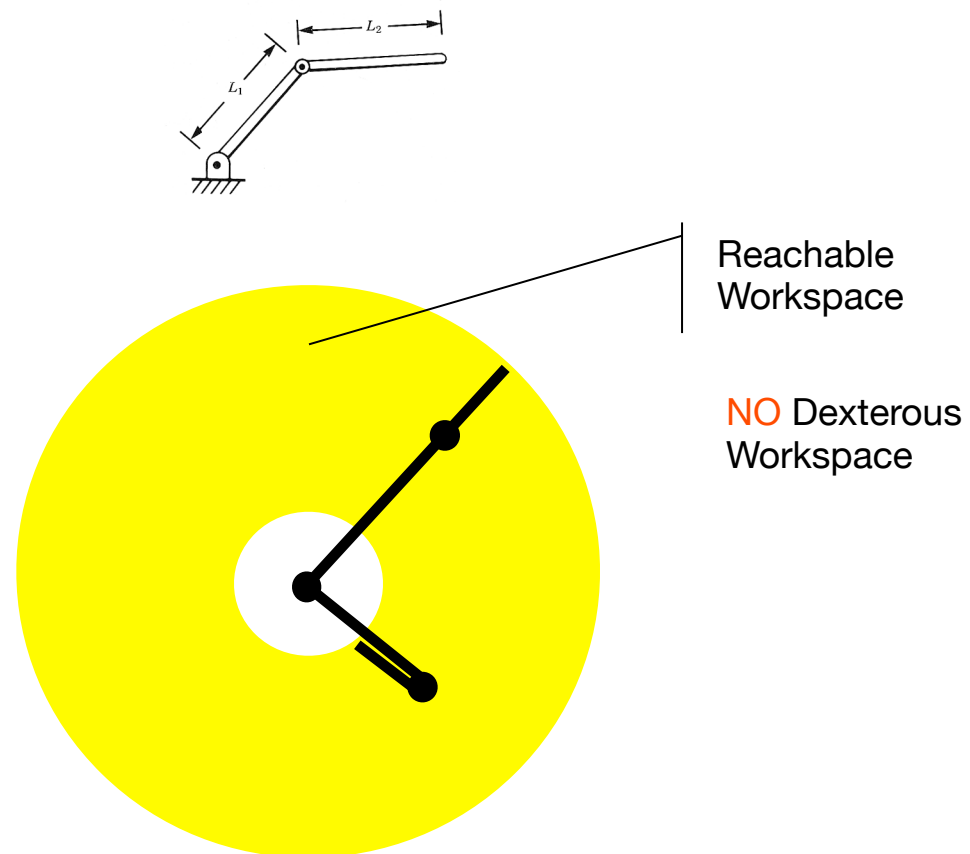
Inverse Manipulator Kinematics

Workspace: 2R Example 1 $L_1 = L_2$



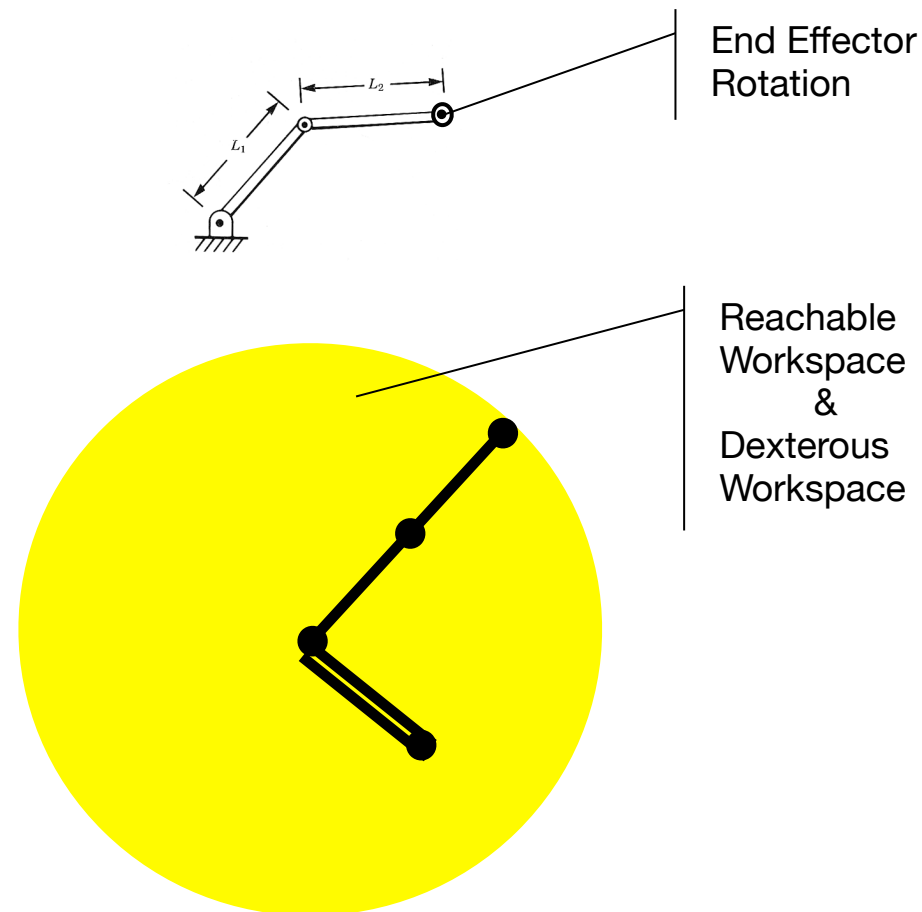
Inverse Manipulator Kinematics

Workspace: 2R Example 2 $L_1 \neq L_2$



Inverse Manipulator Kinematics

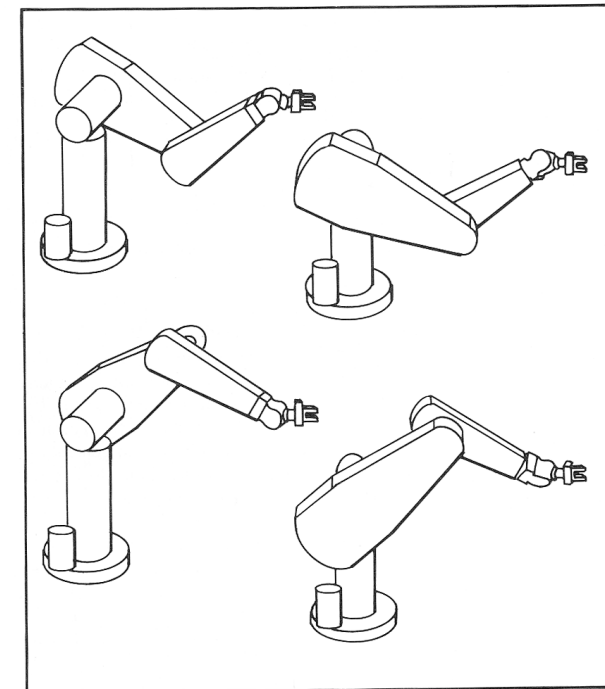
Workspace: 3R Example 3 $L_1 = L_2$



Inverse Manipulator Kinematics

Solvability - Multiple Solution

- Multiple solutions are a common problem that can occur when solving inverse kinematics
- The number of solutions depends on the number of joints in the manipulator but is also a function of the links parameters $a_i \alpha_i d_i \theta_i$
- Example: The PUMA 560 can reach certain goals with 8 different (solutions) arm configurations
 - Four solutions are depicted
 - Four solutions are related to a “flipped” wrist



$$\theta'_4 = \theta_4 + 180^\circ$$

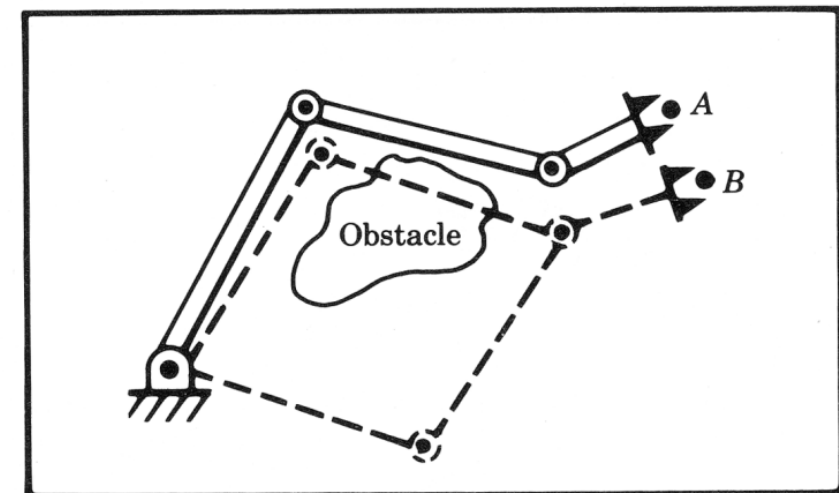
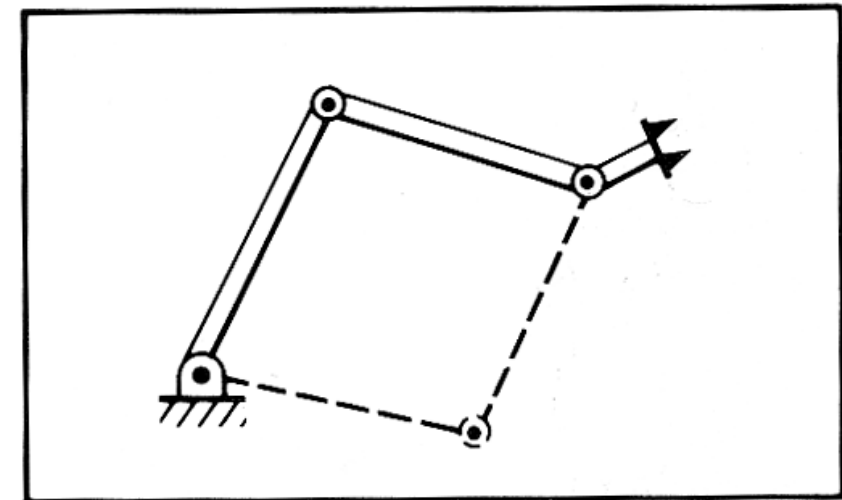
$$\theta'_5 = -\theta_5$$

$$\theta'_6 = \theta_6 + 180^\circ$$

Inverse Manipulator Kinematics

Solvability - Multiple Solution

- **Problem:** The fact that a manipulator has multiple solutions may cause problems because the system has to be able to choose one
- **Solution:** Decision criteria
 - The closest (geometrically) - minimizing the amount that each joint is required to move
 - Note 1: input argument - present position of the manipulator
 - Note 2: Joint Weight - Moving small joints (wrist) instead of moving large joints (Shoulder & Elbow)
 - Obstacles exist in the workspace - avoiding collision



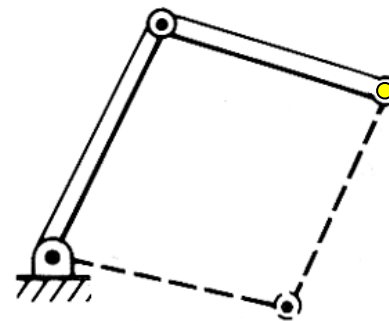
Inverse Manipulator Kinematics

Solvability - Multiple Solution - Number of DOF

Task Definition - Position the end effector in a specific point in the plane (2D)

- No. of DOF = No. of DOF of the task

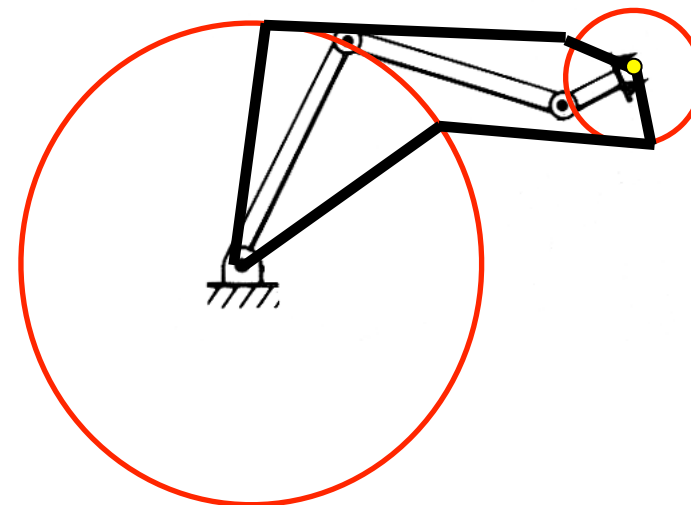
Number of solution: 2
(elbow up/down)



- No. of DOF > No. of DOF of the task

Number of solution: ∞

Self Motion - The robot can be moved without moving the the end effector from the goal



Inverse Manipulator Kinematics

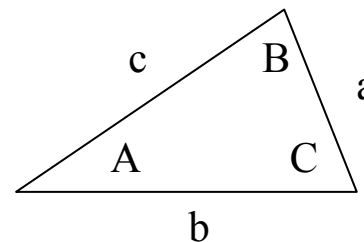
Solvability - Method of Solutions

- **Solution** (Inverse Kinematics)- A “solution” is the set of joint variables associated with an end effector’s desired position and orientation.
- **No general algorithms** that lead to the solution of inverse kinematic equations.
- **Solution Strategies**
 - **Closed form Solutions** - An analytic expression includes all solution sets.
 - **Algebraic Solution** - Trigonometric (Nonlinear) equations
 - **Geometric Solution** - Reduces the larger problem to a series of plane geometry problems.
 - **Numerical Solutions** - Iterative solutions will not be considered in this course.

Inverse Manipulator Kinematics

Mathematical Equations

- Law of Sines / Cosines - For a general triangle



$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

- Sum of Angles

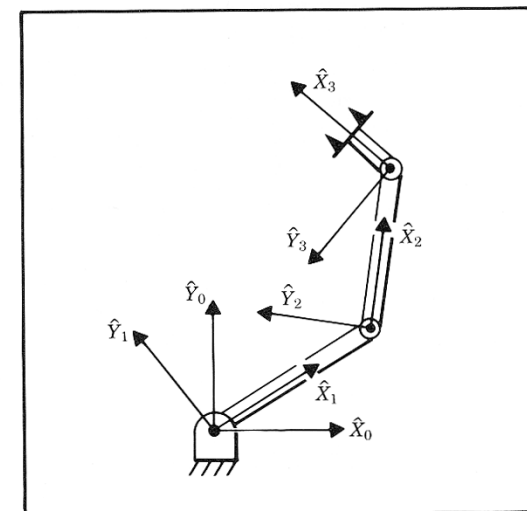
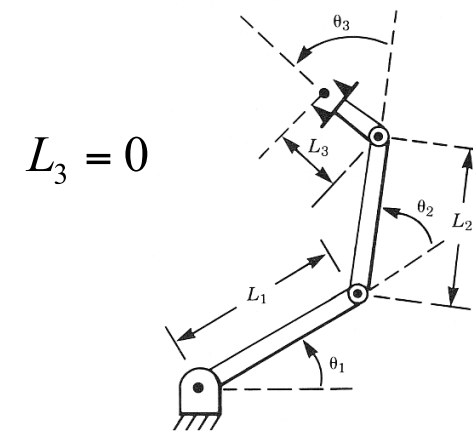
$$\sin(\theta_1 \pm \theta_2) = s_{12} = c_1 s_2 \pm s_1 c_2$$

$$\cos(\theta_1 \pm \theta_2) = c_{12} = c_1 c_2 \mp s_1 s_2$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 1/15

i	$\alpha_i - 1$	$a_i - 1$	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3



Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 2/15

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	L_1	0	θ_2
3	0	L_2	0	θ_3

$${}^0_1T = \begin{bmatrix} c1 & -s1 & 0 & 0 \\ s1 & c1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2T = \begin{bmatrix} c2 & -s2 & 0 & L1 \\ s2 & c2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c3 & -s3 & 0 & L2 \\ s3 & c3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 3/15

$${}^B T = {}^0 T = {}^0 T_1 {}^1 T_2 {}^2 T_3 = \begin{bmatrix} c_1 c_2 c_3 - c_1 s_2 s_3 - s_1 s_2 c_3 - s_1 c_2 s_3 & -c_1 c_2 s_3 - c_1 s_2 c_3 + s_1 s_2 s_3 - s_1 c_2 c_3 & 0 & c_1(L_2 c_2 + L_1) - s_1 s_2 L_2 \\ s_1 c_2 c_3 - s_1 s_2 s_3 + c_1 s_2 c_3 + c_1 c_2 s_3 & -s_1 c_2 s_3 - s_1 s_2 c_3 - c_1 s_2 s_3 + c_1 c_2 c_3 & 0 & s_1(L_2 c_2 + L_1) + c_1 s_2 L_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Using trigonometric identities to simplify ${}^B T$, the solution to the forward kinematics is:

$${}^B T = {}^0 T = {}^0 T_3 = \begin{bmatrix} c_{123} & -s_{123} & 0 & L_1 c_1 + L_2 c_{12} \\ s_{123} & c_{123} & 0 & L_1 s_1 + L_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- where

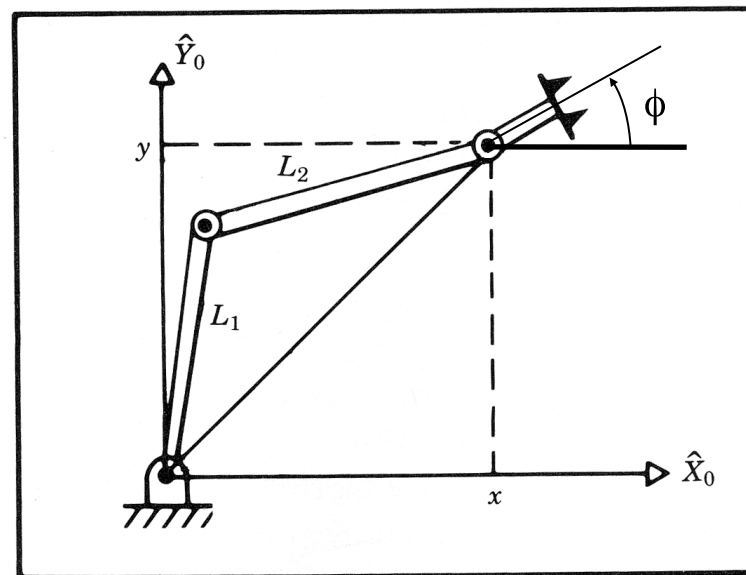
$$c_{123} = \cos(\theta_1 + \theta_2 + \theta_3) \quad s_{123} = \sin(\theta_1 + \theta_2 + \theta_3)$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 4/15

- **Given:**

- **Direct Kinematics:** The homogenous transformation from the base to the wrist B_wT
- **Goal Point Definition:** For a planar manipulator, specifying the goal can be accomplished by specifying three parameters: The position of the wrist in space (x, y) and the orientation of link 3 in the plane relative to the $\hat{X}$ axis (ϕ)



Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 5/15

- **Problem:**

What are the joint angles $(\theta_1, \theta_2, \theta_3)$ as a function of the wrist position and orientation (x, y, ϕ)

- **Solution:**

The goal in terms of position and orientation of the wrist expressed in terms of the homogeneous transformation is defined as follows

$${}^B T_{Goal} = \begin{bmatrix} c_\phi & -s_\phi & 0 & x \\ s_\phi & c_\phi & 0 & y \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}^B T = {}^0 T_3 = \begin{bmatrix} c_{123} & -s_{123} & 0 & L_1 c_1 + L_2 c_{12} \\ s_{123} & c_{123} & 0 & L_1 s_1 + L_2 s_{12} \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 6/15

$${}^B T_{Goal} = {}^0 T_3$$

- A set of four nonlinear equations which must be solved for $\theta_1, \theta_2, \theta_3$

$$c_\phi = c_{123}$$

$$s_\phi = s_{123}$$

$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

- Solving for θ_2
- If we square x and y add them while making use of $c_{12} = c_1 c_2 - s_1 s_2$; $s_{12} = c_1 s_2 + s_1 c_2$ we obtain

$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1 l_2 c_2$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 7/15

$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

$$c_{12} = c_1 c_2 - s_1 s_2$$

$$s_{12} = c_1 s_2 + s_1 c_2$$

$$x^2 + y^2 = l_1^2 + l_2^2 + 2l_1 l_2 c_2$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 8/15

- Solving for c_2 we obtain

$$c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$$

- Note: In order for a solution to exist, the right hand side must have a value between -1 and 1. Physically if this constraints is not satisfied, then the goal point is too far away for the manipulator to reach.
- Assuming the goal is in the workspace, and making use of we write an expression for s_2 as $c_2^2 + s_2^2 = 1$

$$s_2 = \pm\sqrt{1 - c_2^2}$$

- Note: The chose of the sign corresponds to the multiple solutions in which we can choose the “elbow-up” or the “elbow-down” solution

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 9/15

- Finally, we compute θ_2 using the two argument arctangent function

$$\theta_2 = \text{Atan2}(s_2, c_2) = \text{Atan2}(\pm\sqrt{1-c_2^2}, \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2})$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 10/15

- Solving for θ_1
- For solving θ_1 we rewrite the original nonlinear equations using a **change of variables** as follows

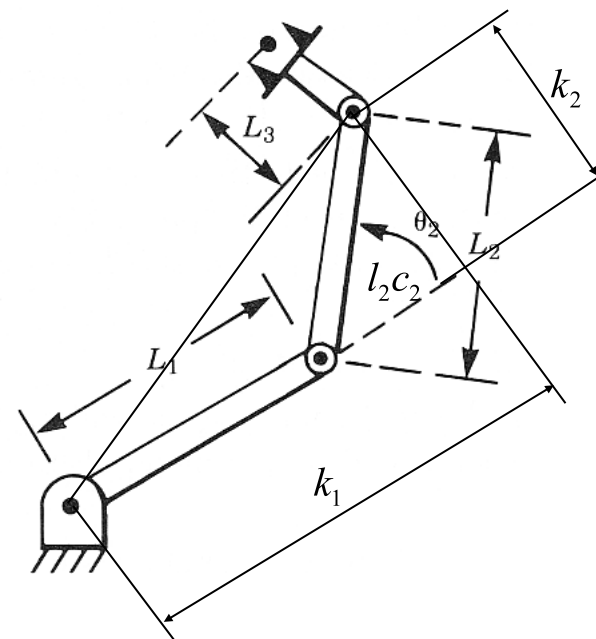
$$x = l_1 c_1 + l_2 c_{12}$$

$$y = l_1 s_1 + l_2 s_{12}$$

$$x = k_1 c_1 - k_2 s_1$$

$$y = k_1 s_1 + k_2 c_1$$

- where
 $k_1 = l_1 + l_2 c_2$
 $k_2 = l_2 s_2$



Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 11/15

Geometrical Interpretation

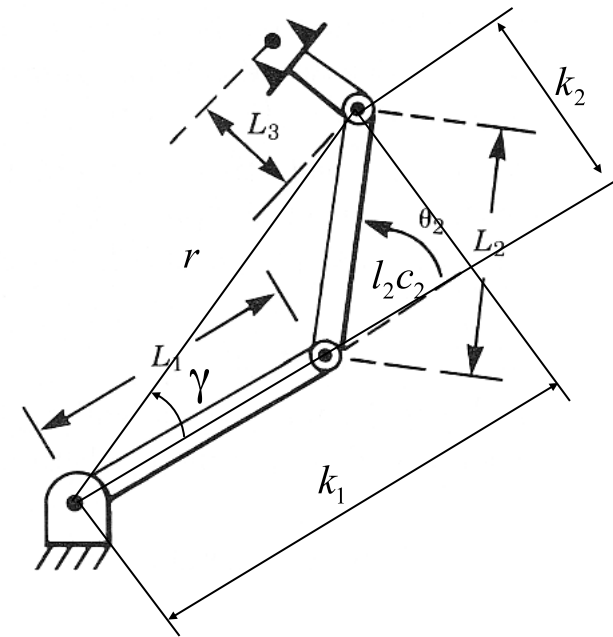
Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 12/15

- Changing the way in which we write the constants k_1 and k_2

$$r = +\sqrt{k_1^2 + k_2^2}$$
$$\gamma = \text{atan2}(k_2, k_1)$$

- Then
$$k_1 = r \cos \gamma$$
$$k_2 = r \sin \gamma$$



Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 13/15

- Base on the previous two transformations the equations can be rewritten as

$$\begin{array}{lll} x = k_1 c_1 - k_2 s_1 & k_1 = r \cos \gamma & x = r \cos \gamma \cos \theta_1 - r \sin \gamma \sin \theta_1 \\ y = k_1 s_1 + k_2 c_1 & k_2 = r \sin \gamma & y = r \cos \gamma \sin \theta_1 + r \sin \gamma \cos \theta_1 \end{array}$$

- or

$$\begin{aligned} \frac{x}{r} &= \cos \gamma \cos \theta_1 - \sin \gamma \sin \theta_1 \\ \frac{y}{r} &= \cos \gamma \sin \theta_1 + \sin \gamma \cos \theta_1 \end{aligned}$$

$$\frac{x}{r} = \cos(\gamma + \theta_1)$$

- or

$$\frac{y}{r} = \sin(\gamma + \theta_1)$$

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 14/15

- Using the two argument arctangent we finally get a solution for θ_1

$$\gamma + \theta_1 = A \tan 2\left(\frac{y}{r}, \frac{x}{r}\right) = A \tan 2(y, x)$$

By Definition

$$\gamma = A \tan 2(k_2, k_1)$$

$$\theta_1 = A \tan 2(y, x) - A \tan 2(k_2, k_1)$$

$$k_1 = l_1 + l_2 c_2$$

$$k_2 = l_2 s_2$$

- Note:

When a choice of a sign is made in the solution of θ_2 ($\theta_2 = A \tan 2(s_2, c_2) = A \tan 2(\pm \sqrt{1 - c_2^2}, \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1 l_2})$), it will cause a sign change in k_2 thus affecting θ_1

Inverse Manipulator Kinematics

Planar RRR (3R): Algebraic Solution 15/15

- Solving for θ_3
- Base on the original equations

$$c_\phi = c_{123}$$

$$s_\phi = s_{123}$$

- We can solve for the sum of $\theta_1, \theta_2, \theta_3$

$$\theta_1 + \theta_2 + \theta_3 = A \tan 2(s_\phi, c_\phi) = \phi$$

$$\theta_3 = \phi - \theta_1 + \theta_2$$

- Note: It is typical with manipulators that have two or more links moving in a plane that in the course of a solution, expressions for sum of joint angles arise

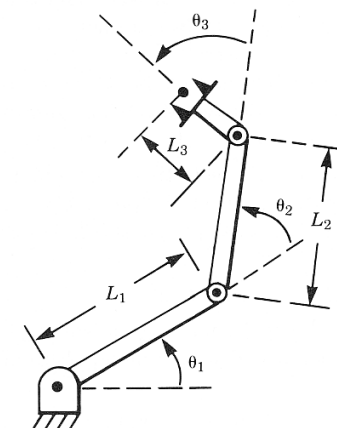
Inverse Manipulator Kinematics

Examples: Geometric Solution, Algebraic Solution

- **Geometric Solution - Concept**

Decompose spatial geometry into several plane geometry

Examples - Planar RRR (3R) manipulators - Geometric Solution

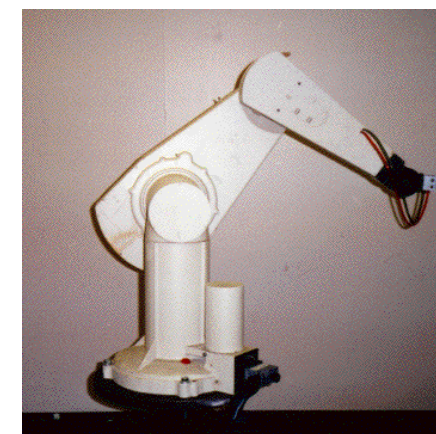


- **Algebraic Solution - Concept**

$${}^0_N T = {}^0_1 T \dots {}^{N-1}_N T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Direct Kinematics Goal (Numeric values)

Examples - PUMA 560 - Algebraic Solution

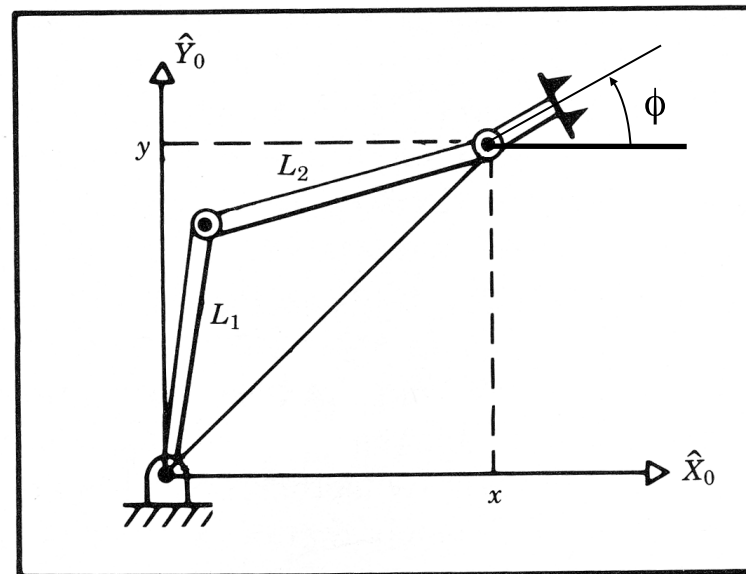


Inverse Manipulator Kinematics

Planar RRR (3R): Geometric Solution 1/4

- **Given:**
 - **Manipulator Geometry**
 - **Goal Point Definition:** The position x, y and orientation ϕ of the wrist in space
- **Problem:**

What are the joint angles ($\theta_1, \theta_2, \theta_3$) as a function of the goal (wrist position and orientation)



Inverse Manipulator Kinematics

Planar RRR (3R): Geometric Solution 2/4

- **Solution:**

- We can apply the law of cosines to solve for θ_2

$$r^2 = x^2 + y^2 = l_1^2 + l_2^2 - 2l_1l_2 \cos(180 - \theta_2)$$

- Since $\cos(180 - \theta_2) = -\cos\theta_2$

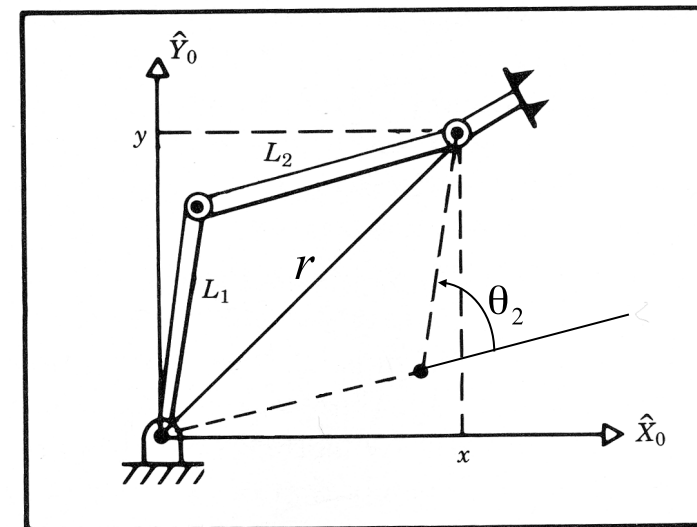
- We have $c_2 = \frac{x^2 + y^2 - l_1^2 - l_2^2}{2l_1l_2}$

$$s_2 = \pm\sqrt{1 - c_2^2}$$

$$\theta_2 = \text{atan2}(s_2, c_2)$$

- Note : Should be checked by the computational algorithm to verify existence of solutions.

$$l_1 + l_2 \geq \sqrt{x^2 + y^2}$$



Inverse Manipulator Kinematics

Planar RRR (3R): Geometric Solution 3/4

- By definition

$$\theta_1 = \beta \pm \psi$$

$\swarrow \theta_2 < 0$
 $\searrow \theta_2 > 0$

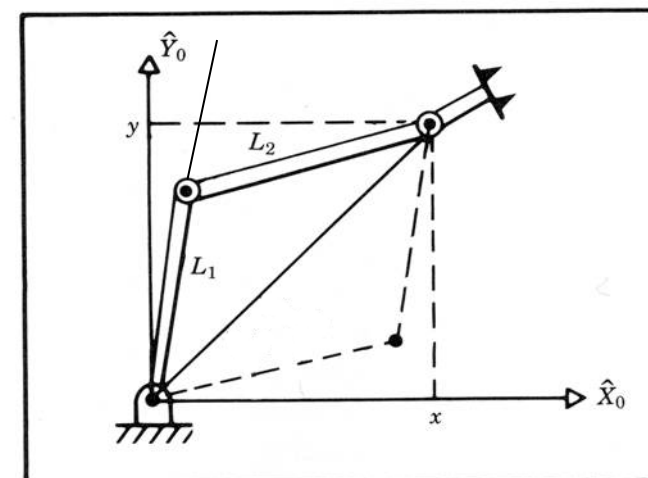
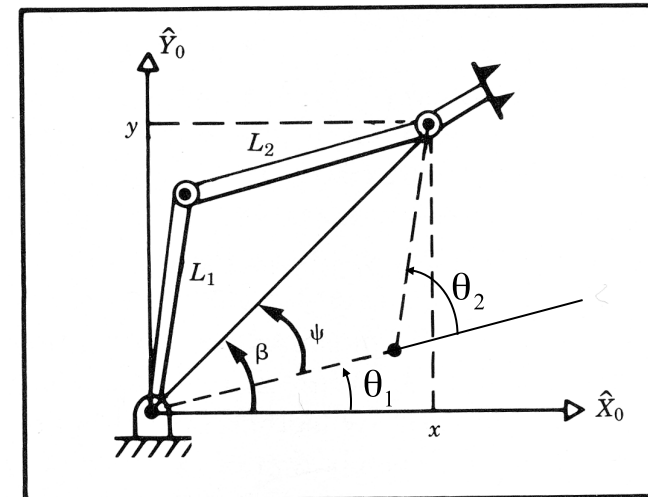
- Defining β as a function of x, y

$$\beta = \text{atan2}(y, x)$$

- Applying the law of cosine to find

$$\cos \psi = \frac{x^2 + y^2 + l_1^2 - l_2^2}{2l_1 \sqrt{x^2 + y^2}}$$

- Note: $0^\circ \leq \psi \leq 180^\circ$



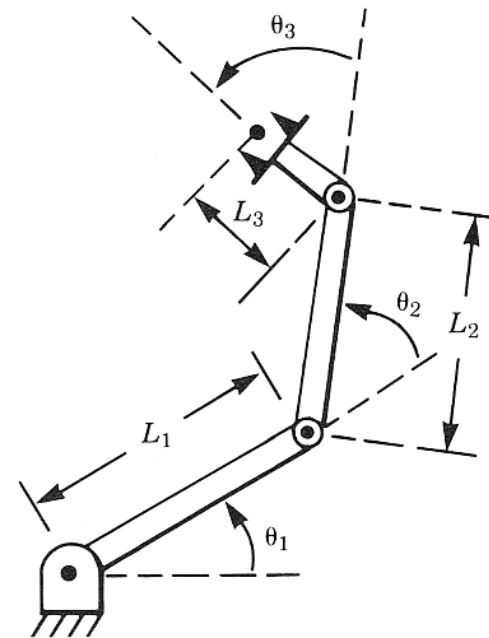
Inverse Manipulator Kinematics

Planar RRR (3R): Geometric Solution 4/4

- Angle in the plane add up to define the orientation of the last link

$$\phi = \theta_1 + \theta_2 + \theta_3$$

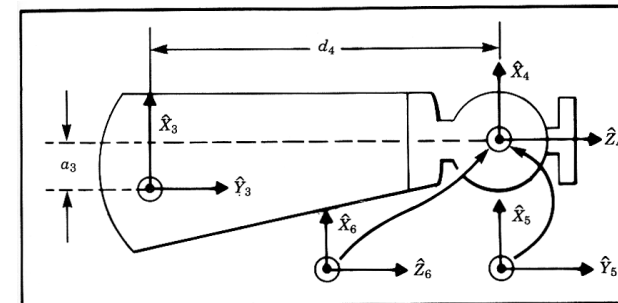
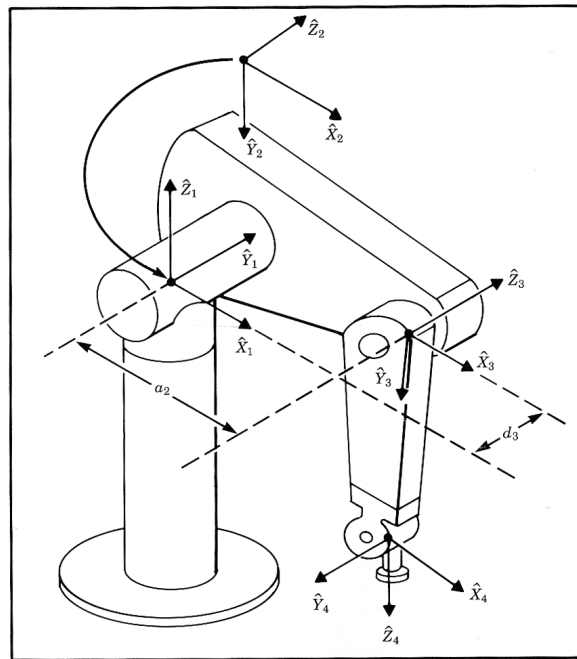
$$\theta_3 = \phi - \theta_1 - \theta_2$$



Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 1

- **Given:**
 - **Direct Kinematics:** The homogenous transformation from the base to the wrist
 - **Goal Point Definition:** The position and orientation of the wrist in space B_wT



Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 2

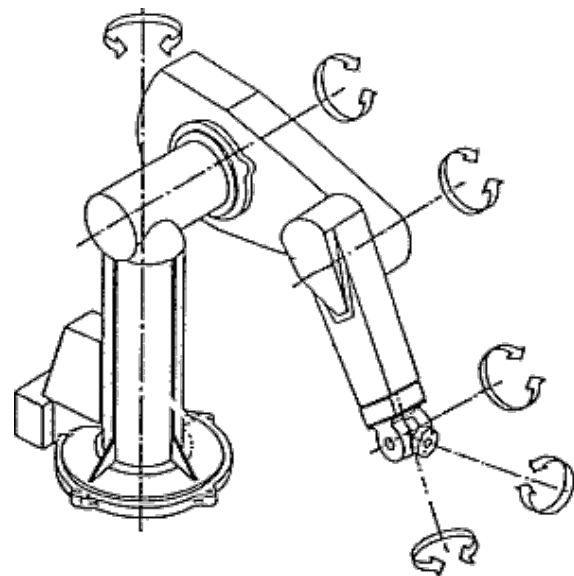
- Problem:**

What are the joint angles ($\theta_1 \cdots \theta_6$) as a function of the wrist position and orientation (or when 0_6T is given as numeric values)

$${}^0_6T = {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6) = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Direct Kinematics

Goal



Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 3

- **Solution (General Technique):** Multiplying each side of the direct kinematics equation by an inverse transformation matrix for separating out variables in search of solvable equation

$${}^0_6T = {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6)$$

- Put the dependence on θ_1 on the left hand side of the equation by multiplying the direct kinematics eq. with $[{}^0_1T(\theta_1)]^{-1}$ gives

$$[{}^0_1T(\theta_1)]^{-1} {}^0_6T = \overset{I}{[{}^0_1T(\theta_1)]^{-1} {}^0_1T(\theta_1)} {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6)$$

- And continue...

$$[{}^0_3T(\theta_1, \theta_2, \theta_3)]^{-1} {}^0_6T = \overset{I}{[{}^0_3T(\theta_1)]^{-1} {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3)} {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6)$$

$$[{}^0_4T(\theta_1, \theta_2, \theta_3, \theta_4)]^{-1} {}^0_6T = \overset{I}{[{}^0_4T(\theta_1, \theta_2, \theta_3, \theta_4)]^{-1} {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4)} {}^4_5T(\theta_5) {}^5_6T(\theta_6)$$

$$[{}^0_5T(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)]^{-1} {}^0_6T = \overset{I}{[{}^0_5T(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)]^{-1} {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5)} {}^5_6T(\theta_6)$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 4

- Put the dependence on θ_1 on the left hand side of the equation by multiplying the direct kinematics eq. with ${}_1^0T(\theta_1)^{-1}$ gives

$${}_1^0T(\theta_1)^{-1} {}_6^0T = I {}_1^0T(\theta_1) {}_2^1T(\theta_2) {}_3^2T(\theta_3) {}_4^3T(\theta_4) {}_5^4T(\theta_5) {}_6^5T(\theta_6)$$

$${}_1^0T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad {}_0^1T = [{}_1^0T]^{-1} = \begin{bmatrix} c\theta_1 & s\theta_1 & 0 & 0 \\ -s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad [{}_B^AT]^{-1} = {}_A^BT = \begin{bmatrix} {}_B^AR^T & -{}_B^AR^T {}_A^BP_{BORG} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} c_1 & s_1 & 0 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}_6^1T$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 5

$${}^1_2T = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_2 & -c\theta_2 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^2_3T = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_2 \\ s\theta_3 & c\theta_3 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^3_4T = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & a_3 \\ 0 & 0 & 1 & d_4 \\ -s\theta_4 & -c\theta_4 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^4_5T = \begin{bmatrix} c\theta_5 & -s\theta_5 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ s\theta_5 & c\theta_5 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^5_6T = \begin{bmatrix} c\theta_6 & -s\theta_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s\theta_6 & -c\theta_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1_6T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^1r_{11} = c_{23} [c_4 c_5 c_6 - s_4 s_6] - s_{23} s_5 s_6,$$

$${}^1r_{21} = -s_4 c_5 c_6 - c_4 s_6,$$

$${}^1r_{31} = -s_{23} [c_4 c_5 c_6 - s_4 s_6] - c_{23} s_5 c_6,$$

$${}^1r_{12} = -c_{23} [c_4 c_5 s_6 + s_4 c_6] + s_{23} s_5 s_6,$$

$${}^1r_{22} = s_4 c_5 s_6 - c_4 c_6,$$

$${}^1r_{32} = s_{23} [c_4 c_5 s_6 + s_4 c_6] + c_{23} s_5 s_6,$$

$${}^1r_{13} = -c_{23} c_4 s_5 - s_{23} c_5,$$

$${}^1r_{23} = s_4 s_5,$$

$${}^1r_{33} = s_{23} c_4 s_5 - c_{23} c_5,$$

$$\rightarrow {}^1p_x = a_2 c_2 + a_3 c_{23} - d_4 s_{23},$$

$$\rightarrow {}^1p_y = d_3,$$

$$\rightarrow {}^1p_z = -a_3 s_{23} - a_2 s_2 - d_4 c_{23}.$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 6

- Solution for

$$\Rightarrow \begin{bmatrix} c_1 & s_1 & 0 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} \stackrel{\downarrow}{=} {}^1_6T$$

- Equating the (2,4) elements from both sides of the equation we have

$$-s_1 p_x + c_1 p_y = d_3$$

How to solve this?

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 7

$$-s_1 p_x + c_1 p_y = d_3$$

- To solve the equation of this form we make the trigonometric substitution

$$\begin{aligned} p_x &= \rho \cos \phi & \rho &= \sqrt{p_x^2 + p_y^2} \\ p_y &= \rho \sin \phi & \phi &= \operatorname{atan2}(p_y, p_x) \end{aligned}$$

- Substituting p_x, p_y with ρ, ϕ we obtain

$$c_1 s_\phi - s_1 c_\phi = \frac{d_3}{\rho}$$

- Using the difference of angles formula

$$\sin(\phi - \theta_1) = \frac{d_3}{\rho}$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 8

- Based on $\sin^2(\phi - \theta_1) + \cos^2(\phi - \theta_1) = 1$

$$\cos(\phi - \theta_1) = \pm \sqrt{1 - \frac{d_3^2}{\rho^2}} \qquad \sin(\phi - \theta_1) = \frac{d_3}{\rho}$$

- and so

$$\phi - \theta_1 = A \tan 2 \left(\frac{d_3}{\rho}, \pm \sqrt{1 - \frac{d_3^2}{\rho^2}} \right)$$

- The solution for θ_1 may be written

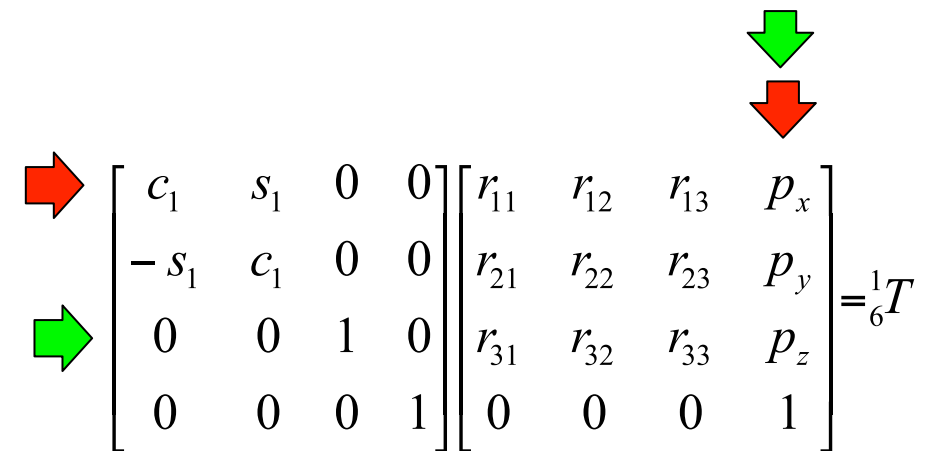
$$\theta_1 = A \tan 2(p_y, p_x) - A \tan 2 \left(\frac{d_3}{\rho}, \pm \sqrt{1 - \frac{d_3^2}{\rho^2}} \right)$$

- Note: we have found two possible solutions for θ_1 corresponding to the +/- sign

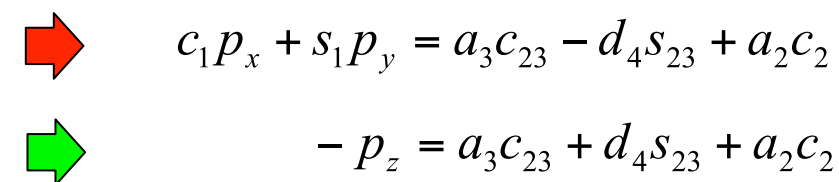
Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 9

- Solution for θ_3
- Equating the (1,4) element and (3,4) element


$$\begin{matrix} \rightarrow \\ \rightarrow \end{matrix} \begin{bmatrix} c_1 & s_1 & 0 & 0 \\ -s_1 & c_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = {}^1_6T$$

- We obtain


$$\begin{aligned} c_1 p_x + s_1 p_y &= a_3 c_{23} - d_4 s_{23} + a_2 c_2 \\ -p_z &= a_3 c_{23} + d_4 s_{23} + a_2 c_2 \end{aligned}$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 10

- If we square the following equations and add the resulting equations

$$\Rightarrow -s_1 p_x + c_1 p_y = d_3$$

$$\Rightarrow c_1 p_x + s_1 p_y = a_3 c_{23} - d_4 s_{23} + a_2 c_2$$

$$\Rightarrow -p_z = a_3 s_{23} + d_4 c_{23} + a_2 s_2$$

Inverse Manipulator Kinematics

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Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 12

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 13

- we obtain $a_3 c_3 - d_4 s_3 = K$

where
$$K = \frac{p_x^2 + p_y^2 + p_z^2 - a_2^2 - a_3^2 - d_3^2 - a_4^2}{2a_2}$$

- Note that the dependence on θ_1 has been removed. Moreover the eq. for θ_3 is of the same form as the eq. for θ_1 and so may be solved by the same kind of trigonometric substitution to yield a solution for θ_3

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 14

$$\theta_3 = A \tan 2(a_3, d_4) - A \tan 2\left(K, \pm \sqrt{a_3^2 + d_4^2 - K^2}\right)$$

- Note that the +/- sign leads to two different solution for θ_3

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 15

- Solution for θ_2
$${}^0_3T(\theta_1, \theta_2, \theta_3)^{-1} {}^0_6T = [{}^0_3T(\theta_1)]^{-1} {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6)$$

$$\begin{bmatrix} c_1 c_{23} & s_1 c_{23} & -s_{23} & -a_2 c_3 \\ -c_1 s_{23} & -s_1 s_{23} & -c_{23} & a_2 s_3 \\ -s_1 & c_1 & 0 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & -c_4 s_5 & a_3 \\ s_5 c_6 & -s_5 s_6 & -c_5 & d_4 \\ -s_4 c_5 c_6 - c_4 s_6 & s_4 c_5 s_6 - c_4 c_6 & s_4 s_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- Equating the (1,4) element and (2,4) element we obtain

$$\begin{aligned} c_1 c_{23} p_x + s_1 c_{23} p_y - s_{23} p_z - a_2 c_3 &= a_3 \\ -c_1 s_{23} p_x - s_1 s_{23} p_y - c_{23} p_z + a_2 s_3 &= d_4 \end{aligned}$$

- These equations may be solved simultaneously for s_{23} and c_{23} resulting in

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 16

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 17

$$s_{23} = \frac{(-a_3 - a_2 c_2) p_z + (c_1 p_x + s_1 p_y)(a_2 s_3 - d_4)}{p_z^2 + (c_1 p_x + s_1 p_y)^2}$$

$$c_{23} = \frac{(a_2 s_3 - d_4) p_z - (-a_3 - a_2 c_3)(c_1 p_x + s_1 p_y)}{p_z^2 + (c_1 p_x + s_1 p_y)^2}$$

- Since the denominator are equal and positive, we solve for the sum of θ_2 and θ_3 as

$$\theta_{23} = A \tan 2 \left[\frac{(-a_3 - a_2 c_2) p_z + (c_1 p_x + s_1 p_y)(a_2 s_3 - d_4)}{(a_2 s_3 - d_4) p_z - (-a_3 - a_2 c_3)(c_1 p_x + s_1 p_y)} \right]$$

- The equation computes four values of θ_{23} according to the four possible combination of solutions for θ_1 and θ_3
- Then, four possible solutions for θ_2 are computed as

$$\theta_2 = \theta_{23} - \theta_3$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 18

- Solution for θ_4
- Equating the (1,3) and the (3,3) elements

$$\begin{array}{c} \Rightarrow \\ \Rightarrow \end{array} \begin{bmatrix} c_1 c_{23} & s_1 c_{23} & -s_{23} & -a_2 c_3 \\ -c_1 s_{23} & -s_1 s_{23} & -c_{23} & a_2 s_3 \\ -s_1 & c_1 & 0 & -d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_4 c_5 c_6 - s_4 s_6 & -c_4 c_5 s_6 - s_4 c_6 & -c_4 s_5 & a_3 \\ s_5 c_6 & -s_5 s_6 & -c_5 & d_4 \\ -s_4 c_5 c_6 - c_4 s_6 & s_4 c_5 s_6 - c_4 c_6 & s_4 s_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- we get

$$\begin{array}{c} \Rightarrow \\ \Rightarrow \end{array} \begin{array}{l} r_{13} c_1 c_{23} + r_{23} s_1 c_{23} - s_{23} r_{33} = -c_4 s_5 \\ -r_{13} s_1 + r_{23} c_1 = s_4 s_5 \end{array} \quad \Rightarrow \begin{array}{l} c_4 = \frac{-r_{13} c_1 c_{23} - r_{23} s_1 c_{23} + s_{23} r_{33}}{s_5} \\ s_4 = \frac{-r_{13} s_1 + r_{23} c_1}{s_5} \end{array}$$

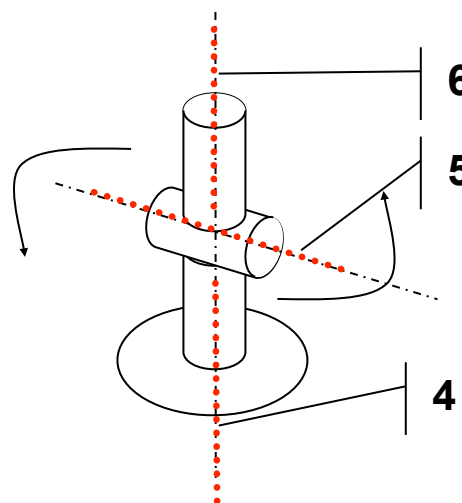
Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 19

- As long as $s_5 \neq 0$ we can solve for θ_4

$$\theta_4 = \text{Atan2}(-r_{13}s_1 + r_{23}c_1, -r_{13}c_1c_{23} - r_{23}s_1c_{23} + s_{23}r_{33})$$

- When $\theta_5 = 0$ the manipulator is in a **singular configuration** in which joint axes 4 and 6 line up and cause the same motion of the last link of the robot. In this case all that can be solved for is the sum or difference of θ_4 and θ_6 . If so θ_4 is chosen arbitrary (usually chosen to be equal to the present value of joint 4), and θ_6 is computed later.



Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 20

- Solution for θ_5

$$[{}^0_4T(\theta_1, \theta_2, \theta_3, \theta_4)]^{-1} {}^0_6T = [{}^0_4T(\theta_1, \theta_2, \theta_3, \theta_4)]^{-1} {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6)$$

$$\begin{array}{c} \rightarrow \\ \rightarrow \end{array} \begin{bmatrix} c_1c_{23}c_4 + s_1s_4 & s_1c_{23}c_4 - c_1s_4 & -s_{23}c_4 & -a_2c_3c_4 + d_3s_4 - a_3c_4 \\ -c_1c_{23}s_4 + s_1c_4 & -s_1c_{23}s_4 - c_1c_4 & s_{23}s_4 & a_2c_3s_4 + d_3c_4 - a_3s_4 \\ -c_1s_{23} & -s_1s_{23} & c_{23} & a_2s_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_5c_6 & -c_5s_6 & -s_5 & 0 \\ s_6 & c_6 & 0 & 0 \\ s_5c_6 & -s_5s_6 & c_5 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{c} \leftarrow \\ \leftarrow \end{array}$$

- Equating the (1,3) and the (3,3) elements we get

$$\rightarrow r_{13}(c_1c_{23}c_4 + s_1s_4) + r_{23}(s_1c_{23}c_4 - c_1s_4) - r_{33}(s_{23}c_4) = -s_5$$

$$\rightarrow r_{13}(-c_1s_{23}) + r_{23}(-s_1s_{23}) + r_{33}(-c_{23}) = c_5$$

- We can solve for θ_5

$$\theta_5 = \text{Atan2}(s_5, c_5)$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 21

- Solution for θ_6

$$[{}^0_5T(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)]^{-1} {}^0_6T = [{}^0_5T(\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)]^{-1} {}^0_1T(\theta_1) {}^1_2T(\theta_2) {}^2_3T(\theta_3) {}^3_4T(\theta_4) {}^4_5T(\theta_5) {}^5_6T(\theta_6)$$

$$\begin{array}{c} \text{Orange Arrow} \\ \text{Blue Arrow} \end{array} \begin{bmatrix} c_1c_{23}c_4c_5 + s_1s_4c_5 - s_1c_{23}c_4s_5 + c_1s_4s_5 & s_{23}c_4 & c_1c_{23}c_4s_5 + s_1s_4s_5 + s_1c_{23}c_4c_5 - c_1s_4c_5 & -a_2c_3c_4 + d_3s_4 - a_3c_4 \\ -c_1c_{23}s_4c_5 + s_1c_4c_5 + s_1c_{23}s_4s_5 + c_1c_4s_5 & -s_{23}s_4 & -c_1c_{23}s_4s_5 + s_1c_4s_5 - s_1c_{23}s_4c_5 - c_1c_4c_5 & a_2c_3s_4 + d_3c_4 - a_3s_4 \\ -c_1s_{23}c_5 + s_1s_{23}s_5 & -c_{23} & -c_1s_{23}s_5 - s_1s_{23}c_5 & a_2s_3 - d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{c} \text{Orange Arrow} \downarrow \text{Blue Arrow} \\ \text{Orange Arrow} \downarrow \text{Blue Arrow} \end{array} \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{array}{c} \text{Orange Arrow} \downarrow \text{Blue Arrow} \\ \text{Orange Arrow} \downarrow \text{Blue Arrow} \end{array} \begin{bmatrix} c_6 & -s_6 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -s_6 & -c_6 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{array}{c} \text{Orange Arrow} \\ \text{Blue Arrow} \end{array}$$

- Equating the (3,1) and the (1,1) elements we get

$$\begin{array}{c} \text{Blue Arrow} \\ \text{Orange Arrow} \end{array} \begin{array}{l} r_{11}(c_1s_{23}c_5 - s_1s_{23}s_5) + r_{21}(c_{23}) + r_{31}(+c_1s_{23}s_5 + s_1s_{23}c_5) = s_6 \\ r_{11}(c_1c_{23}c_4c_5 + s_1s_4c_5 - s_1c_{23}c_4s_5 + c_1s_4s_5) + r_{21}(s_{23}c_4) + r_{31}(c_1c_{23}c_4s_5 + s_1s_4s_5 + s_1c_{23}c_4c_5 - c_1s_4c_5) = c_6 \end{array}$$

- We can solve for θ_6

$$\theta_6 = \text{Atan2}(s_6, c_6)$$

Inverse Manipulator Kinematics

PUMA 560: Algebraic Solution 22

Summary - Number of Solutions

- Four solution

$$\theta_1 = A \tan 2(p_y, p_x) - A \tan 2\left(\frac{d_3}{\rho}, \pm \sqrt{1 - \frac{d_3^2}{\rho^2}}\right)$$

$$\theta_3 = A \tan 2(a_3, d_4) - A \tan 2\left(K, \pm \sqrt{a_3^2 + d_4^2 - K^2}\right)$$

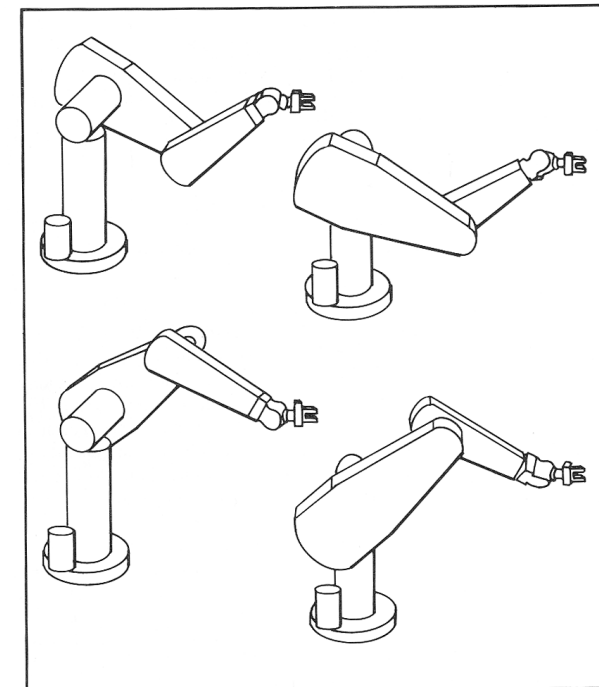
- For each of the four solutions the wrist can be flipped

$$\theta_4' = \theta_4 + 180^\circ$$

$$\theta_5' = -\theta_5$$

$$\theta_6' = \theta_6 + 180^\circ$$

- After all eight solutions have been computed, some or all of them may have to be discarded because of joint limit violations.
- Of the remaining valid solutions, usually the one closest to the present manipulator configuration is chosen.



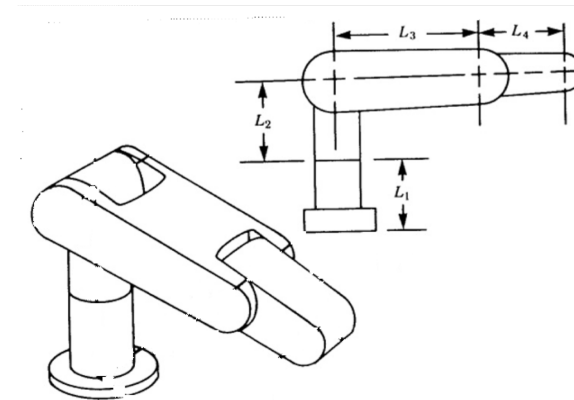
Inverse Manipulator Kinematics

Examples: Geometric Solution, Algebraic Solution

- **Geometric Solution - Concept**

Decompose spatial geometry into several plane geometry

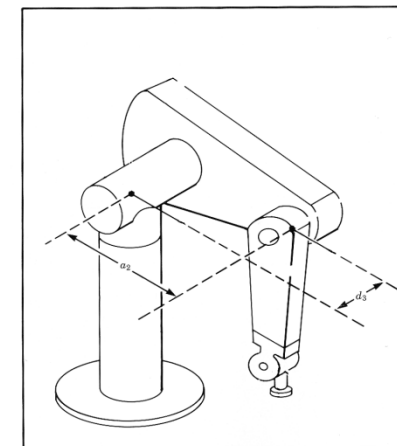
Example - 3D RRR (3R) manipulators -
Geometric Solution



- **Algebraic Solution (closed form)**

Piepers Method - Last three consecutive axes intersect at one point

Example - PUMA 560

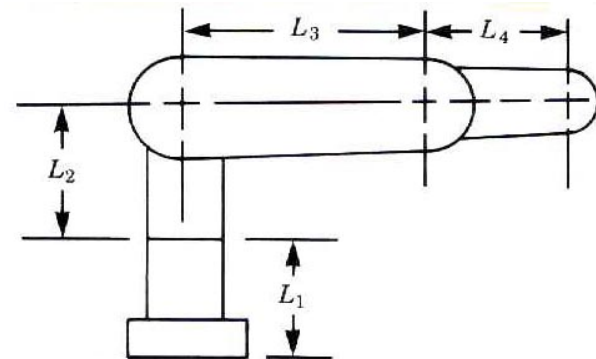
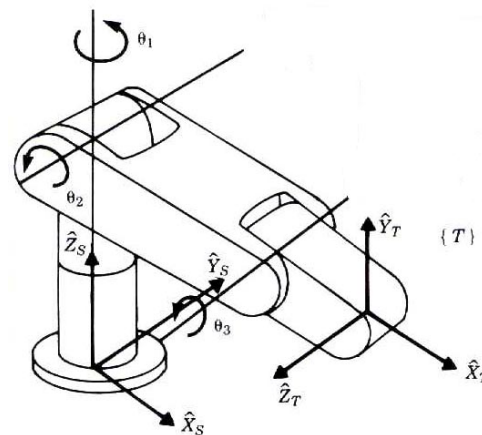


Inverse Manipulator Kinematics

3D RRR (3R): Geometric Solution 1/6

- **Given:**

- **Manipulator Geometry**
- **Goal Point Definition:** The position x_d, y_d, z_d of the wrist in space



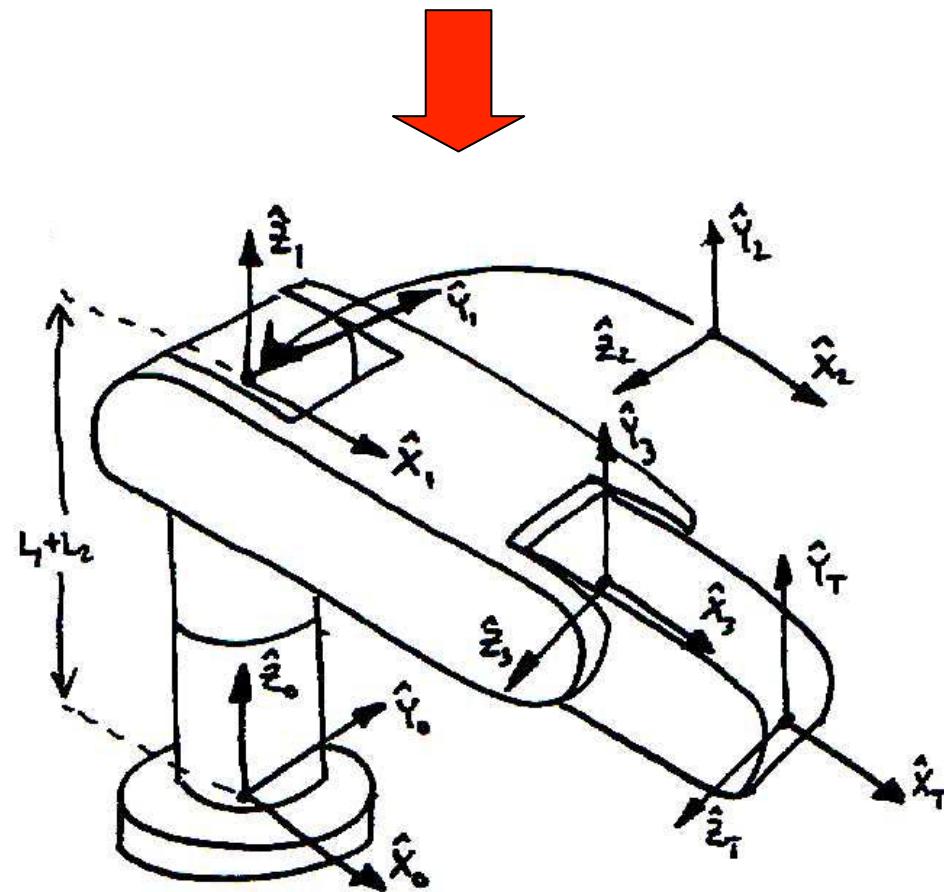
- **Problem:**

What are the joint angles $(\theta_1, \theta_2, \theta_3)$ as a function of the goal (wrist position and orientation)

Inverse Manipulator Kinematics

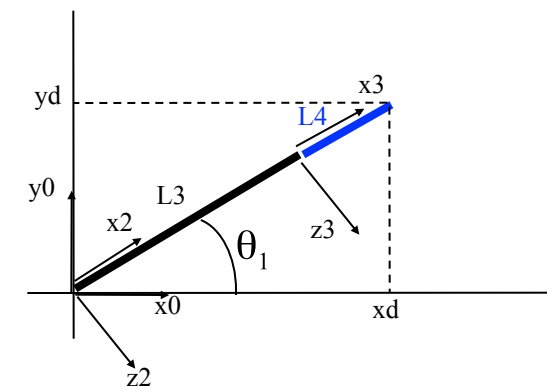
3D RRR (3R): Geometric Solution 2/6

- The planar geometry - top view of the robot



$$\{0\} = \{S\}$$

$$\{4\} = \{T\}$$



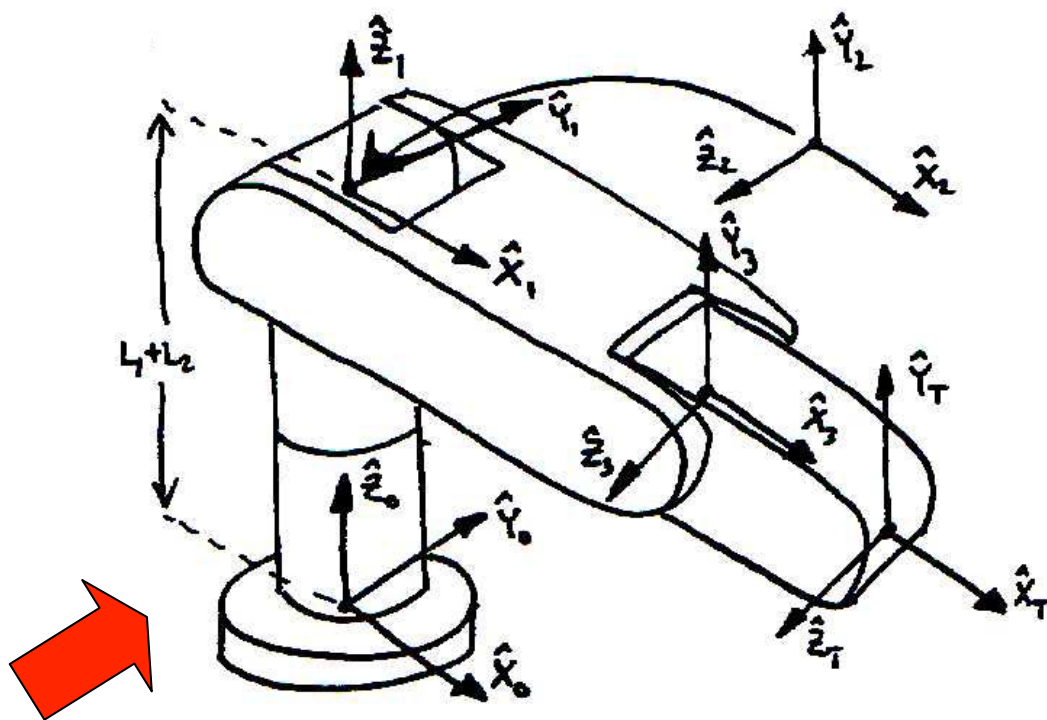
$$\theta_1 = \text{atan2}(y_d, x_d)$$

$$r_1 = \sqrt{x_d^2 + y_d^2}$$

Inverse Manipulator Kinematics

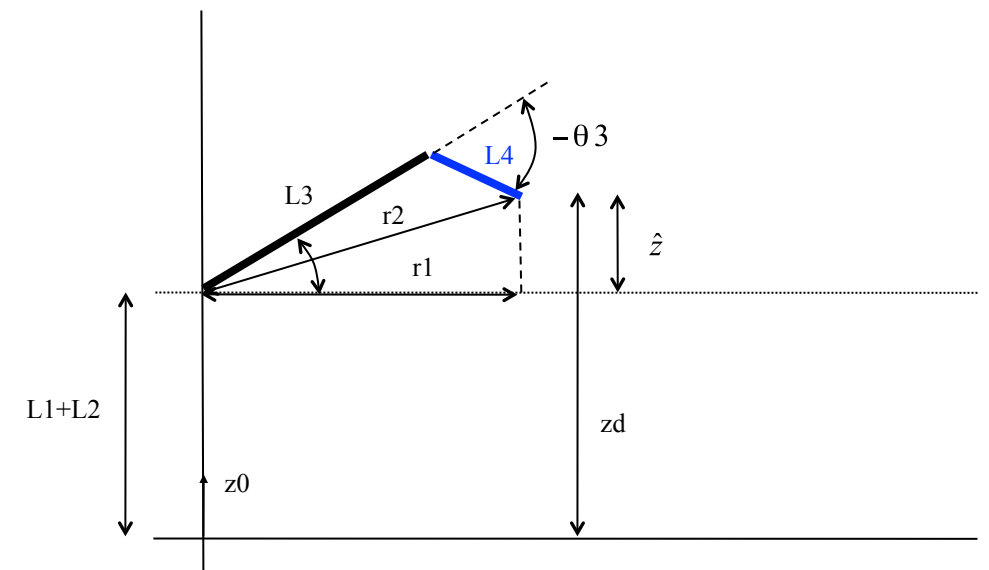
3D RRR (3R): Geometric Solution 3/6

- The planar geometry - side view of the robot



$$\{0\} = \{S\}$$

$$\{4\} = \{T\}$$



$$r_2 = \sqrt{r_1^2 + \hat{z}^2} = \sqrt{x_d^2 + y_d^2 + \hat{z}^2} = \sqrt{x_d^2 + y_d^2 + (z_d - (L_1 + L_2))^2}$$

$$\text{where } \hat{z} = z_d - (L_1 + L_2)$$

Inverse Manipulator Kinematics

3D RRR (3R): Geometric Solution 4/6

- By Apply the law of cosines we get

$$r_2^2 = L_3^2 + L_4^2 - 2L_3L_4 \cos(180 + \theta_3) = L_3^2 + L_4^2 + 2L_3L_4 \cos(\theta_3)$$

- Rearranging gives

$$c_3 = \frac{r_2^2 - (L_3^2 + L_4^2)}{2L_3L_4}$$

- and

$$s_3 = \pm \sqrt{1 - c_3^2}$$

- Solving for θ_3 we get

$$\theta_3 = \text{Atan}(\pm \sqrt{1 - c_3^2}, c_3)$$

where c_3 is defined above in terms of known parameters L_3, L_4, x_d, y_d , and z_d

Inverse Manipulator Kinematics

3D RRR (3R): Geometric Solution 5/6

- Finally we need to solve for θ_2

$$\theta_2 = \alpha + \beta$$

$$\alpha = A \tan 2(\hat{z}, r_1)$$

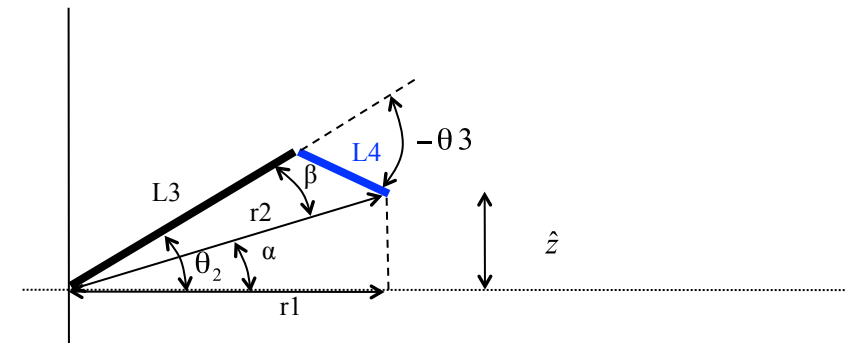
$$\text{where } r_1 = \sqrt{x_d^2 + y_d^2} \quad \hat{z} = z_d - (L_1 + L_2)$$

- Based on the law of cosines we can solve for β

$$L_4^2 = r_2^2 + L_3^2 - 2r_2L_3 \cos(\beta)$$

$$c_\beta = \frac{r_2^2 + L_3^2 - L_4^2}{2r_2L_3}$$

$$\beta = A \tan 2(\pm \sqrt{1 - c_\beta^2}, c_\beta)$$



$$\theta_2 = A \tan 2(z_d - (L_1 + L_2), \sqrt{x_d^2 + y_d^2}) + A \tan 2(\pm \sqrt{1 - c_\beta^2}, c_\beta)$$

Inverse Manipulator Kinematics

3D RRR (3R): Geometric Solution 6/6

- Summary

$$\theta_1 = A \tan 2(y_d, x_d)$$

$$\theta_2 = A \tan 2(z_d - (L_1 + L_2), \sqrt{x_d^2 + y_d^2}) +$$

$$A \tan 2\left(\pm \sqrt{1 - \left(\frac{x_d^2 + y_d^2 + (z_d - (L_1 + L_2))^2 + L_3^2 - L_4^2}{2\sqrt{x_d^2 + y_d^2 + (z_d - (L_1 + L_2))L_3}^2}\right)^2}, \frac{x_d^2 + y_d^2 + (z_d - (L_1 + L_2))^2 + L_3^2 - L_4^2}{2\sqrt{x_d^2 + y_d^2 + (z_d - (L_1 + L_2))L_3}^2}\right)$$

$$\theta_3 = A \tan\left(\pm \sqrt{1 - \left[\frac{x_d^2 + y_d^2 + (z_d - (L_1 + L_2))^2 - (L_3^2 + L_4^2)}{2L_3L_4}\right]^2}, \frac{x_d^2 + y_d^2 + (z_d - (L_1 + L_2))^2 - (L_3^2 + L_4^2)}{2L_3L_4}\right)$$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 1

- Equation $\sin \theta = a \quad a \in [-1, 1]$
 $\cos \theta = b \quad b \in [-1, 1]$
- Solution (Unique) $\theta = A \tan 2(a, b)$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 2

- Equation

$$\sin\theta = a \quad a \in [-1, 1]$$

$$\cos\theta = \pm\sqrt{1-a^2}$$

$$\cos\theta = b \quad b \in [-1, 1]$$

$$\sin\theta = \pm\sqrt{1-b^2}$$

- Solution

$$\theta = A \tan 2(a, \pm\sqrt{1-a^2})$$

$$\theta = A \tan 2(\pm\sqrt{1-b^2}, b)$$

- Two Solutions

$$\theta_1 = \theta$$

$$\theta_2 = 180 - \theta$$

$$\theta_1 = \theta$$

$$\theta_2 = -\theta$$

- Singularity at the Boundary

$$\text{When } \theta = \pm 90^\circ, \quad |a| = 1$$

$$\text{When } \theta = 0^\circ, 180^\circ, \quad |b| = 1$$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 3

- Equation $a(\cos\theta) + b(\sin\theta) = 0$

$$\frac{\sin\theta}{\cos\theta} = -\frac{a}{b}$$

- Solution

- Two Solutions 180° apart

$$\theta = A \tan 2(a, -b)$$

$$\theta = A \tan 2(-a, b)$$

- Singularity $b = 0$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 4

- Equation

$$a(\cos\theta) + b(\sin\theta) = c$$

$$a, b, c \neq 0$$

- Solution

- Two Solutions

$$\theta = A \tan 2(\pm \sqrt{a^2 + b^2 - c^2}, c) + A \tan 2(b, a)$$

- For a solution to exist

$$a^2 + b^2 - c^2 > 0$$

- No solution (outside of the workspace)

$$a^2 + b^2 - c^2 < 0$$

- One solution (singularity)

$$a^2 + b^2 - c^2 = 0$$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 4

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 4

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 5

- Equation

$$a \cos \theta + c \sin \theta = d$$

$$e \cos \theta + f \sin \theta = g$$

- For a solution to exist, the determinant must not be zero

$$af - ce \neq 0$$

- Solution

$$\begin{cases} \sin \theta = \frac{ag - de}{af - ce} \\ \cos \theta = \frac{df - cg}{af - ce} \end{cases} \Rightarrow \theta = \begin{cases} \text{atan2}(ag - de, df - cg), & af - ce > 0 \\ \text{atan2}(de - ag, cg - df), & af - ce < 0 \end{cases}$$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 6

- Transcendental equations are difficult to solve because they are a function of $c\theta, s\theta$

$$f(c\theta, s\theta) = k$$

- Making the following substitutions yields an expression in terms of a single veritable u , Using this substitutions, transcendental equation are converted into polynomial equation

$$u = \tan \frac{\theta}{2}$$

$$\cos\theta = \frac{1-u^2}{1+u^2}$$

$$\sin\theta = \frac{2u}{1+u^2}$$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 6

- Transcendental equation

$$ac\theta + bs\theta = c$$

- Substitute $c\theta, s\theta$ with the following equations

$$\cos\theta = \frac{1-u^2}{1+u^2}$$

$$\sin\theta = \frac{2u}{1+u^2}$$

- yields

$$a(1-u^2) + 2bu = c(1+u^2)$$

$$(a+c)u^2 - 2bu + (c-a) = 0$$

Inverse Manipulator Kinematics

Some Inverse Kinematics Formulas: Case 6

- Which is solved by the quadratic formula to be

$$u = \frac{b \pm \sqrt{b^2 - a^2 - c^2}}{a + c}$$

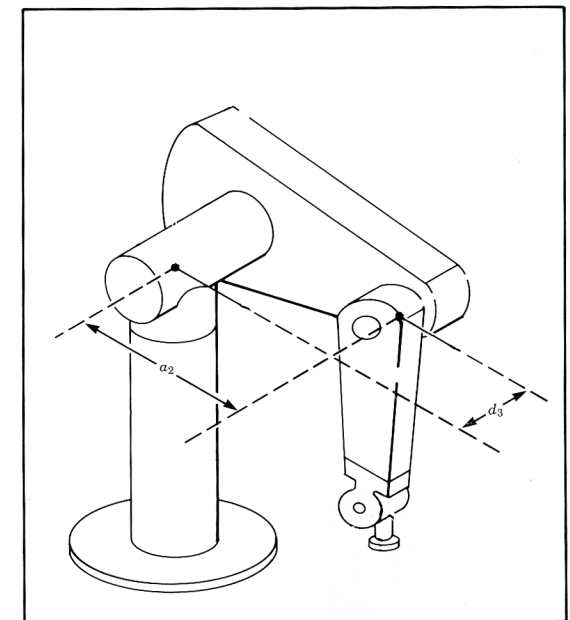
$$\theta = 2 \tan^{-1} \left(\frac{b \pm \sqrt{b^2 - a^2 - c^2}}{a + c} \right)$$

- Note
 - If U is complex there is no real solution to the original transcendental equation
 - If $a + c = 0$ then $\theta = 180^\circ$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

- **Inverse Kinematics - Close Form Solution (Sufficient Conditions) – 6 DOF**
 - Three consecutive axes intersect at a single point
 - Three consecutive revolute joints are parallel (meet at a point at infinity)
- **Pieper's Solution** - Closed form solution for a serial 6 DOF
- Pieper's method applies to the majority of commercially available industrial robots
- Example: (Puma 560)
 - All 6 joints are revolute joints
 - The last 3 joints are intersecting



Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

- **Given:**

- **Manipulator Geometry:** 6 DOF & DH parameters
 - All 6 joints are revolute joints
 - The last 3 joints are intersecting
- **Goal Point Definition:** The position and orientation of the wrist in space

$${}^0_6T = {}^0_1T {}^1_2T {}^2_3T {}^3_4T {}^4_5T {}^5_6T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & p_x \\ r_{21} & r_{22} & r_{23} & p_y \\ r_{31} & r_{32} & r_{33} & p_z \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- **Problem:**

What are the joint angles ($\theta_1, \theta_2, \theta_3, \theta_4, \theta_5, \theta_6$) as a function of the goal (wrist position and orientation)

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

- When the last three axes of a 6 DOF robot intersect, the origins of link frame {4}, {5}, and {6} are all located at the point of intersection. This point is given in the base coordinate system as

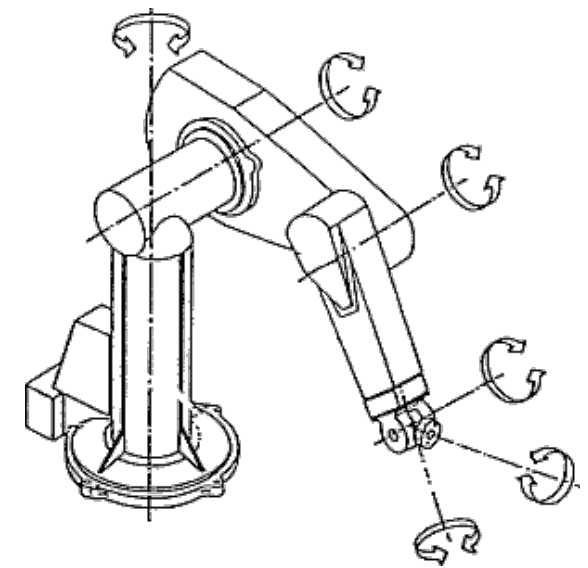
$${}^0P_{4org} = {}^0T_1 {}^1T_2 {}^2T_3 P_{4org}$$

- From the general forward kinematics method for determining homogeneous transforms using DH parameters, we know:

$${}^{i-1}T_i = \begin{bmatrix} {}^{i-1}R_i & {}^{i-1}P_{iorg} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ s\theta_i c\alpha_{i-1} & c\theta_i c\alpha_{i-1} & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\theta_i s\alpha_{i-1} & c\theta_i s\alpha_{i-1} & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

- For $i=4$

$${}^3T_4 = \begin{bmatrix} {}^3R_4 & {}^3P_{4org} \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} c\theta_4 & -s\theta_4 & 0 & a_3 \\ s\theta_4 c\alpha_3 & c\theta_4 c\alpha_3 & -s\alpha_3 & -s\alpha_3d_4 \\ s\theta_4 s\alpha_3 & c\theta_4 s\alpha_3 & c\alpha_3 & c\alpha_3d_4 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

- Using the fourth column and substituting for ${}^3P_{4org}$, we find

$${}^0P_{4org} = {}^0T_1 {}^1T_2 {}^2T_3 {}^3P_{4org} = {}^0T_1 {}^1T_2 {}^2T_3 \begin{bmatrix} a_3 \\ -s\alpha_3 d_4 \\ c\alpha_3 d_4 \\ 1 \end{bmatrix} \quad {}^0P_{4org} = {}^0T_1 {}^1T_2 {}^2T_3 {}^3P_{4org} = {}^0T_1 {}^2T \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$

where

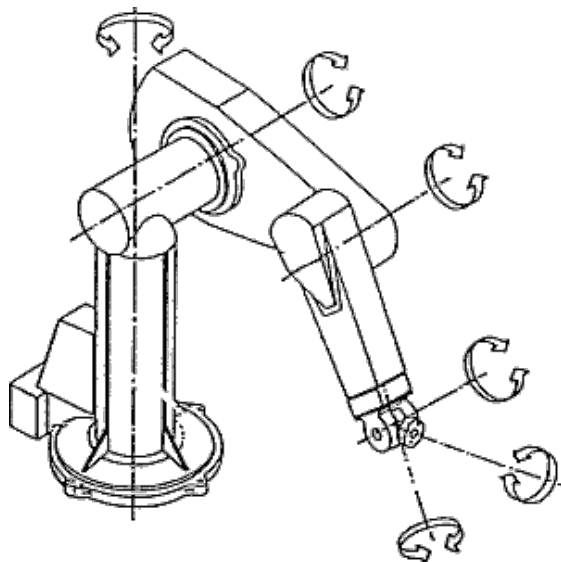
$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^2_3T \begin{bmatrix} a_3 \\ -s\alpha_3 d_4 \\ c\alpha_3 d_4 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = \begin{bmatrix} c\theta_3 & -s\theta_3 & 0 & a_2 \\ s\theta_3 c\alpha_2 & c\theta_3 c\alpha_2 & -s\alpha_2 & -s\alpha_2 d_3 \\ s\theta_3 s\alpha_2 & c\theta_3 s\alpha_2 & c\alpha_2 & c\alpha_2 d_3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_3 \\ -s\alpha_3 d_4 \\ c\alpha_3 d_4 \\ 1 \end{bmatrix}$$

$$f_1(\theta_3) = a_3 c_3 + d_4 s\alpha_3 s_3 + a_2$$

$$f_2(\theta_3) = a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2$$

$$f_3(\theta_3) = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$



Inverse Manipulator Kinematics

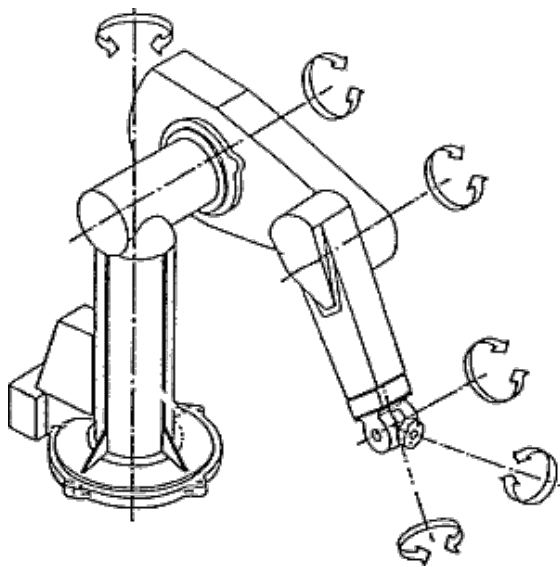
Pieper's Solution - Three Consecutive Axes Intersect

- Repeating the same process again

$${}^0P_{4org} = {}^0T_1 {}^1T_2 {}^2T_3 P_{4org} = {}^0T_1 {}^1T_2 \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix} = {}^0T_1 \begin{bmatrix} g_1(\theta_2) \\ g_2(\theta_2) \\ g_3(\theta_2) \\ 1 \end{bmatrix}$$

where

$$\begin{bmatrix} g_1(\theta_2) \\ g_2(\theta_2) \\ g_3(\theta_2) \\ 1 \end{bmatrix} = {}^1T_2 \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$



$$\begin{bmatrix} g_1(\theta_2) \\ g_2(\theta_2) \\ g_3(\theta_2) \\ 1 \end{bmatrix} = \begin{bmatrix} c\theta_2 & -s\theta_2 & 0 & a_1 \\ s\theta_2 c\alpha_1 & c\theta_2 c\alpha_1 & -s\alpha_1 & -s\alpha_1 d_2 \\ s\theta_2 s\alpha_1 & c\theta_2 s\alpha_1 & c\alpha_1 & c\alpha_1 d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} f_1(\theta_3) \\ f_2(\theta_3) \\ f_3(\theta_3) \\ 1 \end{bmatrix}$$

$$g_1(\theta_2) = c_2 f_1 + s_2 f_2 + a_1$$

$$g_2(\theta_2) = s_2 s\alpha_1 f_1 + c_2 c\alpha_1 f_2 - s\alpha_1 f_3 - d_2 s\alpha_1$$

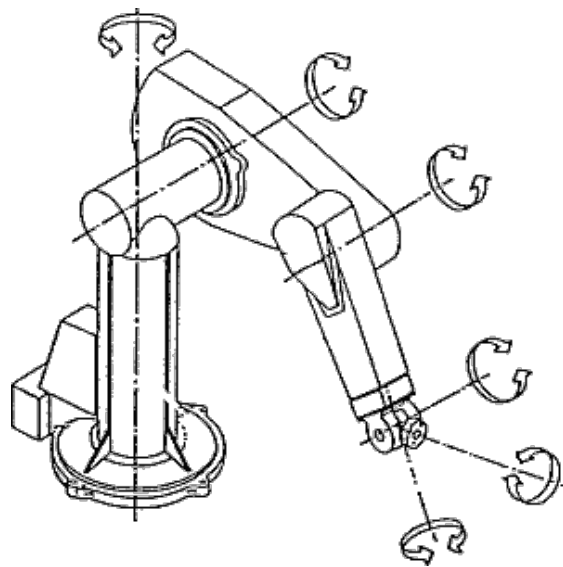
$$g_3(\theta_2) = s_2 s\alpha_1 f_1 + c_2 s\alpha_1 f_2 + c\alpha_1 f_3 + d_2 c\alpha_1$$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

- Repeating the same process for the last time

$${}^0P_{4org} = {}^0T_1 {}^1T_2 {}^2T_3 P_{4org} = {}^0T_1 \begin{bmatrix} g_1(\theta_2) \\ g_2(\theta_2) \\ g_2(\theta_2) \\ 1 \end{bmatrix}$$



$${}^0P_{4org} = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & a_0 \\ s\theta_1 c\alpha_0 & c\theta_1 c\alpha_0 & -s\alpha_0 & -s\alpha_0 d_1 \\ s\theta_1 s\alpha_0 & c\theta_1 s\alpha_0 & c\alpha_0 & c\alpha_0 d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} g_1(\theta_2) \\ g_2(\theta_2) \\ g_3(\theta_2) \\ 1 \end{bmatrix}$$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

- Frame {0} - The frame attached to the base of the robot or link 0 called frame {0}. This frame does not move and for the problem of arm kinematics can be considered as the reference frame.
- Assign {0} to match {1} when the first joint veritable is zero $\theta_1 \neq 0 \quad \alpha_0 = d_1 = a_0 = 0$

$${}^0P_{4org} = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & a_0 \\ s\theta_1 c\alpha_0 & c\theta_1 c\alpha_0 & -s\alpha_0 & -s\alpha_0 d_1 \\ s\theta_1 s\alpha_0 & c\theta_1 s\alpha_0 & c\alpha_0 & c\alpha_0 d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} g_1(\theta_2) \\ g_2(\theta_2) \\ g_3(\theta_2) \\ 1 \end{bmatrix}$$

$$\theta_1 \neq 0 \quad \alpha_0 = d_1 = a_0 = 0$$

$${}^0P_{4org} = \begin{bmatrix} c_1 g_1 - s_1 g_2 \\ s_1 g_1 + c_1 g_2 \\ g_3 \\ 1 \end{bmatrix}$$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

- Through algebraic manipulation of these equations, we can solve for the desired joint angles ($\theta_1, \theta_2, \theta_3$).
- The first step is to square the magnitude of the distance from the frame {0} origin to frame {4} origin.

$$r^2 = \left({}^0P_{4orgx} \right)^2 + \left({}^0P_{4orgy} \right)^2 + \left({}^0P_{4orgz} \right)^2 = g_1^2 + g_2^2 + g_3^2$$

- Using the previously define function for g_i we have

$$r^2 = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2f_3 + a_1(c_2f_1 - s_2f_2)$$

$$Z = {}^0P_{4orgz} = g_3 = s_2s\alpha_1f_1 + c_2s\alpha_1f_2 + c\alpha_1f_3 + d_2c\alpha_1$$

$$g_1(\theta_2) = c_2f_1 + s_2f_2 + a_1$$

$$g_2(\theta_2) = s_2s\alpha_1f_1 + c_2c\alpha_1f_2 - s\alpha_1f_3 - d_2s\alpha_1$$

$$g_3(\theta_2) = s_2s\alpha_1f_1 + c_2s\alpha_1f_2 + c\alpha_1f_3 + d_2c\alpha_1$$

- Applying a substitution of temporary variables, we can write the magnitude squared term along with the z-component of the {0} frame origin to the {4} frame origin distance.

$$r^2 = (k_1c_2 + k_2s_2)2a_1 + k_3$$

$$Z = (k_1s_2 - k_2c_2)s\alpha_1 + k_4$$

with $k_1 = f_1$

$$k_2 = -f_2$$

$$k_3 = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2f_3$$

$$k_4 = f_3c\alpha_1 + d_2c\alpha_1$$

- These equations are useful because dependence on θ_1 has been eliminated, and dependence on θ_2 takes a simple form

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

Consider 3 cases for θ_3

1. CASE 1: $a_1 = 0$

$$\begin{aligned} r^2 &= (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \\ Z &= (k_1 s_2 - k_2 c_2) s\alpha_1 + k_4 \end{aligned}$$

$$r^2 = k_3$$

$$k_3 = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3$$

$$f_1(\theta_3) = a_3 c_3 + d_4 s\alpha_3 s_3 + a_2$$

$$f_2(\theta_3) = a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2$$

$$f_3(\theta_3) = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$

- Solution Methodology - Reduction to Polynomial => Quadratic Equation

$$u = \tan \frac{\theta}{2} \quad \cos \theta = \frac{1 - u^2}{1 + u^2} \quad \sin \theta = \frac{2u}{1 + u^2}$$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

Consider 3 cases for θ_3

2. CASE 2: $s\alpha_1 = 0$

$$r^2 = (k_1 c_2 + k_2 s_2) 2a_1 + k_3$$

$$Z = (k_1 s_2 - k_2 c_2) s\alpha_1 + k_4$$

$$Z = k_4$$

$$k_4 = f_3 c\alpha_1 + d_2 c\alpha_1$$

$$f_3(\theta_3) = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$

- Solution Methodology - Reduction to Polynomial => Quadratic Equation

$$u = \tan \frac{\theta}{2} \quad \cos \theta = \frac{1 - u^2}{1 + u^2} \quad \sin \theta = \frac{2u}{1 + u^2}$$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

Consider 3 cases for θ_3

3. CASE 3: General case

- We can find θ_3 through the following algebraic manipulation:

$$\begin{aligned} r^2 &= (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \\ Z &= (k_1 s_2 - k_2 c_2) s\alpha_1 + k_4 \end{aligned}$$

$$\frac{r^2 - k_3}{2a_1} = (k_1 c_2 + k_2 s_2)$$

$$\frac{Z - k_4}{s\alpha_1} = (k_1 s_2 - k_2 c_2)$$

- squaring both sides, we find

$$\left(\frac{r^2 - k_3}{2a_1} \right)^2 = (k_1 c_2 + k_2 s_2)^2 = k_1^2 c_2^2 + k_2^2 s_2^2 + 2k_1 k_2 c_2 s_2$$

$$\left(\frac{Z - k_4}{s\alpha_1} \right)^2 = (k_1 s_2 - k_2 c_2)^2 = k_1^2 s_2^2 + k_2^2 c_2^2 - 2k_1 k_2 c_2 s_2$$

- Adding these two together and simplifying using the trigonometry identity (Reduction to Polynomial), we find a fourth order equation for θ_3

$$\left(\frac{r^2 - k_3}{2a_1} \right)^2 + \left(\frac{Z - k_4}{s\alpha_1} \right)^2 = k_1^2 + k_2^2$$

$$\begin{aligned} k_1 &= f_1 \\ k_2 &= -f_2 \\ k_3 &= f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3 \\ k_4 &= f_3 c\alpha_1 + d_2 c\alpha_1 \\ f_1(\theta_3) &= a_3 c_3 + d_4 s\alpha_3 s_3 + a_2 \\ f_2(\theta_3) &= a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2 \\ f_3(\theta_3) &= a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2 \end{aligned}$$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

Solve for θ_2

- With θ_3 solved, substitute into r^2, Z to find θ_2

$$\begin{aligned} r^2 &= (k_1 c_2 + k_2 s_2) 2a_1 + k_3 \\ Z &= (k_1 s_2 - k_2 c_2) s\alpha_1 + k_4 \end{aligned}$$

$$k_1 = f_1$$

$$k_2 = -f_2$$

$$k_3 = f_1^2 + f_2^2 + f_3^2 + a_1^2 + d_2^2 + 2d_2 f_3$$

$$k_4 = f_3 c\alpha_1 + d_2 c\alpha_1$$

$$f_1(\theta_3) = a_3 c_3 + d_4 s\alpha_3 s_3 + a_2$$

$$f_2(\theta_3) = a_3 c\alpha_2 s_3 - d_4 s\alpha_3 c\alpha_2 c_3 - d_4 s\alpha_2 c\alpha_3 - d_3 s\alpha_2$$

$$f_3(\theta_3) = a_3 s\alpha_2 s_3 - d_4 s\alpha_3 s\alpha_2 c_3 + d_4 c\alpha_2 c\alpha_3 + d_3 c\alpha_2$$

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

Solve for

- With θ_2, θ_3 solved, substitute into ${}^0P_{4org}$ to find

$${}^0P_{4org} = \begin{bmatrix} c_1 g_1 - s_1 g_2 \\ s_1 g_1 + c_1 g_2 \\ g_3 \\ 1 \end{bmatrix}$$

$${}^0P_{4orgx} = c_1 g_1 - s_1 g_2$$

$${}^0P_{4orgy} = s_1 g_1 + c_1 g_2$$

$$g_1(\theta_2) = c_2 f_1 + s_2 f_2 + a_1$$

$$g_2(\theta_2) = s_2 s \alpha_1 f_1 + c_2 c \alpha_1 f_2 - s \alpha_1 f_3 - d_2 s \alpha_1$$

$$g_3(\theta_2) = s_2 s \alpha_1 f_1 + c_2 s \alpha_1 f_2 + c \alpha_1 f_3 + d_2 c \alpha_1$$

$$f_1(\theta_3) = a_3 c_3 + d_4 s \alpha_3 s_3 + a_2$$

$$f_2(\theta_3) = a_3 c \alpha_2 s_3 - d_4 s \alpha_3 c \alpha_2 c_3 - d_4 s \alpha_2 c \alpha_3 - d_3 s \alpha_2$$

$$f_3(\theta_3) = a_3 s \alpha_2 s_3 - d_4 s \alpha_3 s \alpha_2 c_3 + d_4 c \alpha_2 c \alpha_3 + d_3 c \alpha_2$$

- Solve for θ_1 using the reduction to polynomial method

Inverse Manipulator Kinematics

Pieper's Solution - Three Consecutive Axes Intersect

Solve for $\theta_4, \theta_5, \theta_6$

- Since the last three axes intersect these joint angles affect the orientation of only the last link. We can compute them based only upon the rotation portion of the specified goal 0_6R

$${}^0_6R = {}^0_4R \Big|_{\theta_4=0} {}^4_6R$$

$${}^4_6R \Big|_{\theta_4=0} = {}^0_4R^{-1} \Big|_{\theta_4=0} {}^0_6R$$

${}^0_4R \Big|_{\theta_4=0}$ - The orientation of link frame {4} relative to the base frame {0} when $\theta_4 = 0$

$\theta_4, \theta_5, \theta_6$ are the Euler angles applied to ${}^4_6R \Big|_{\theta_4=0}$

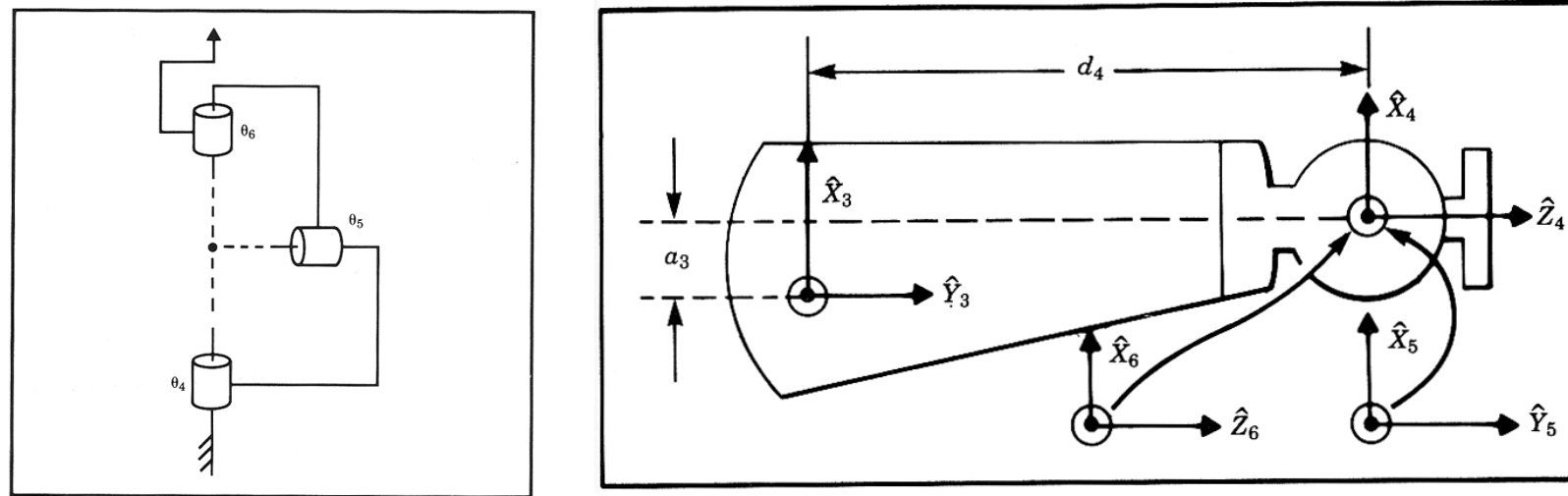
Confused?

See next example for PUMA

Inverse Manipulator Kinematics

Pieper's Solution - Example for PUMA Wrist

- Consider a 3 DOF (wrist) non-planar robot whose axes all intersect at a point.



- Mapping - Rotated Frames - Z-Y-Z Euler Angles

Start with frame {4}.

- Rotate frame {4} about $\hat{Z}_4$ by an angle α
 - Rotate frame {4} about $\hat{Y}_B$ by an angle β
 - Rotate frame {4} about $\hat{Z}_4$ by an angle γ
- } **Euler Angles**

Note: This is not the “Z-Y-X Euler Angles”. See Text Ch 2.

Note: Each rotation is performed about an axis of the *moving reference frame*

Inverse Manipulator Kinematics

Pieper's Solution - Example for PUMA Wrist

$$R_{Z'Y'Z'}(\alpha, \beta, \gamma) = R_Z(\alpha)R_Y(\beta)R_Z(\gamma) = \begin{bmatrix} c\alpha & -s\alpha & 0 \\ s\alpha & c\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \begin{bmatrix} c\gamma & -s\gamma & 0 \\ s\gamma & c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$R_{Z'Y'Z'}(\alpha, \beta, \gamma) = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}$$

$${}^A_B R_{X'Y'Z'}(\alpha, \beta, \gamma) = {}^4_6 R_{\theta_4=0}$$

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}$$

Goal

Direct Kinematics

Inverse Manipulator Kinematics

Pieper's Solution - Example for PUMA Wrist

- Solve for β using element r_{31}, r_{32}, r_{33}

$$\begin{aligned} r_{31} &= -s\beta c\gamma & r_{31}^2 + r_{32}^2 &= s\beta^2 (c\alpha^2 + s\alpha^2) \\ r_{32} &= s\beta s\gamma \\ r_{33} &= c\beta & s\beta &= \pm \sqrt{r_{31}^2 + r_{32}^2} \end{aligned}$$

- Using the Atan2 function, we find

$$\beta = \text{Atan2}\left(\pm \sqrt{r_{31}^2 + r_{32}^2}, r_{33}\right)$$

- Solve for α using elements r_{23}, r_{13}

$$\begin{aligned} r_{13} &= c\alpha s\beta \\ r_{23} &= s\alpha s\beta \\ \alpha &= \text{Atan2}(r_{23} / s\beta, r_{13} / s\beta) \end{aligned}$$

- Solve for γ using elements r_{32}, r_{31}

$$\begin{aligned} r_{32} &= s\beta s\gamma \\ r_{31} &= -s\beta c\gamma \\ \gamma &= \text{Atan2}(r_{32} / s\beta, -r_{31} / s\beta) \end{aligned}$$

Inverse Manipulator Kinematics

Pieper's Solution - Example for PUMA Wrist

- Note: Two answers exist for angle β which will result in two answers each for angles α and γ .

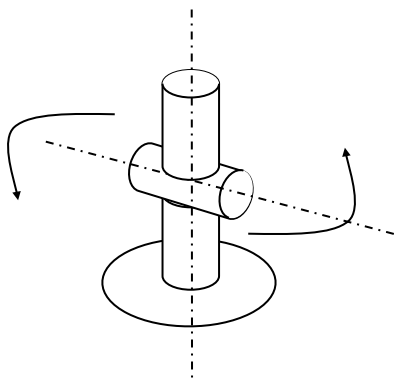
$$\beta = \text{Atan2}\left(\pm \sqrt{r_{31}^2 + r_{32}^2}, r_{33}\right)$$

$$\alpha = \text{Atan2}(r_{23} / s\beta, r_{13} / s\beta)$$

$$\gamma = \text{Atan2}(r_{32} / s\beta, -r_{31} / s\beta)$$

Note: The text says only compute the Beta angle between 0° and 180°

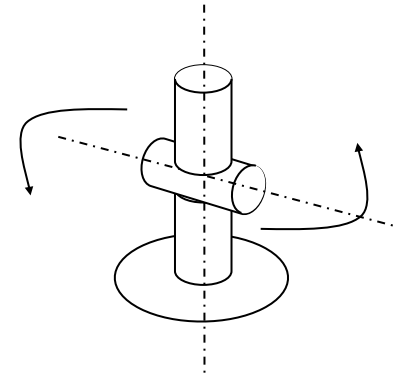
- If $\beta = 0^\circ, \beta = 180^\circ \Rightarrow s\beta = 0$ the solution degenerates



Inverse Manipulator Kinematics

Pieper's Solution - Example for PUMA Wrist

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c\alpha c\beta c\gamma - s\alpha s\gamma & -c\alpha c\beta s\gamma - s\alpha c\gamma & c\alpha s\beta \\ s\alpha c\beta c\gamma + c\alpha s\gamma & -s\alpha c\beta s\gamma + c\alpha c\gamma & s\alpha s\beta \\ -s\beta c\gamma & s\beta s\gamma & c\beta \end{bmatrix}$$



$$\beta = 0^\circ$$

$$\begin{bmatrix} r_{11} & r_{12} & r_{13} \\ r_{21} & r_{22} & r_{23} \\ r_{31} & r_{32} & r_{33} \end{bmatrix} = \begin{bmatrix} c\alpha c\gamma - s\alpha s\gamma & -c\alpha s\gamma - s\alpha c\gamma & 0 \\ s\alpha c\gamma + c\alpha s\gamma & -s\alpha s\gamma + c\alpha c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c(\alpha + \gamma) & -s(\alpha + \gamma) & 0 \\ s(\alpha + \gamma) & c(\alpha + \gamma) & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- We are left with $(\gamma + \alpha)$ for every case. This means we can't solve for either, just their sum.

Inverse Manipulator Kinematics

Pieper's Solution - Example for PUMA Wrist

- One possible convention is to choose $\alpha = 0^\circ$

- The solution can be calculated to be

$$\beta = 0$$

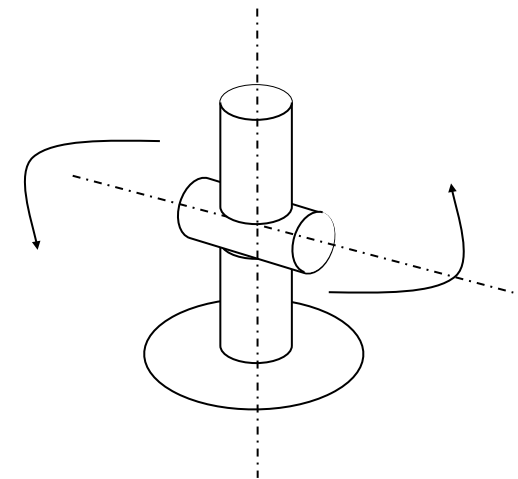
$$\alpha = 0$$

$$\gamma = \text{Atan2}(-r_{12}, r_{11}) = \text{Atan2}(s\gamma, c\gamma)$$

$$\beta = 180$$

$$\alpha = 0$$

$$\gamma = \text{Atan2}(r_{12}, -r_{11}) = \text{Atan2}(s\gamma, c\gamma)$$



- For this example, the singular case results in the capability for self-rotation. That is, the middle link can rotate while the end effector's orientation never changes.